Fractal Routing Is Not Weight Tying

A matched-budget test of scale-recursive routing against flat top-2 MoE routing at ≤4M parameters

Companion to `research/PAPER_1_matched_cost.md` and `research/FINDINGS_AND_THEORY.md`. Those documents stand unchanged; their subject was one cell applied repeatedly along depth, and their conclusion — weight tying buys parameters, not compute, and adaptive depth has no headroom at matched quality on this task — remains correct within that scope. This document tests a different object: routing structure that recurs across **scale** rather than across **step**.

Budget note. Every number here comes from a run trained on `data/calc_mix` for the number of steps in its caption (9,000 unless stated), one shared optimizer/schedule, seed 0. Cost is MACs per **item**, using calc_mix's own measured mean sequence length (44.31 prompt+target tokens from `data/calc_mix/test.meta.jsonl`, n=5,000). Accuracy is best-by-exact-match for every family — one checkpoint rule for both sides. Every table is emitted by `scripts/hier_tables.py`; if a number is not printed there it does not belong in this document.

Abstract. The architecture proposed in the earlier work was *transformers within transformers*: a router at level *n* whose experts are themselves routed cells, self-similar across **scale**. What was tested there was one cell applied eight times — self-similar across **step** — and its negative result does not transfer. Here we build the nested architecture (`src/hmoe/hier.py`) and compare it to a flat top-2 MoE at matched total parameters, matched active parameters per token and matched MACs/item (all three within 0.7%), on calc_mix, 9,000 and 18,000 steps, two seeds. **Result: the hypothesis is not supported; the robust finding is that content-dependent routing costs accuracy in both families.** Against the best flat model available the nested model is statistically indistinguishable at seed 0 (−0.6 points, McNemar p = 0.72) and *significantly worse* at seed 1 (−4.6, p = 2.3e-05), so **no claim of superiority over a well-configured flat MoE survives replication**; the 18k-step seed-0 win (+6.4, p = 0.0011) also inverts at seed 1 (−1.8, p = 0.18). What does replicate at both seeds is narrower: nesting beats any flat model *forced to the same fine granularity*. On the paired 500-item test split (McNemar), matched G=8 (8 leaves of width 31 per outer level, same active width units, same MACs) gives **+8.2 points at seed 0 (61.00% vs 52.80%, p = 1.1e-08) and +3.0 at seed 1 (57.60% vs 54.60%, p = 0.015)**; matched G=16 gives **+8.0 (63.20% vs 55.20%, p = 1.9e-07) and +9.6 (57.60% vs 48.00%, p = 2.4e-09)**. On the 1,000-item val split the same contrasts are +6.7/+3.7 (G=8) and +5.9/+8.8 (G=16). The flat model loses 4.7–6.6 points as its pool gets finer (val 58.50% at G=2 → 57.00% at G=4 → 51.90% at G=8, partially recovering to 53.80% at G=16); the nested model gains across the same range. Nesting is therefore **granularity tolerance**, not an efficiency win. Meanwhile **every routing-prediction measurement is negative**: difficulty-conditioned selection divergence is ≤ 0.0011 nats at both scales (sub-router *less* difficulty-aware than its parent), no specialisation by operator or digit position, and settling depth 7.1–8.0 of 8 outer levels with no separation between easy and hard items — there is no early settling and no hierarchical specialisation. What survives is the structural claim: cross-scale update coherence is **0.093** (and 0.082 across inner sub-depth), far from the ≈ 0.85 of the weight-tied loop, so the scales genuinely do different work. The mechanism question is closed by reciprocal controls at bit-identical parameter and MAC counts: replacing routing with a **fixed random k-subset average** raises nested accuracy from 58.60% to 61.60% val (66.00% vs 61.00% test, p = 1.2e-04) — and raises *flat* accuracy by **more**: on the paired test split +3.2 points at G=2 (61.60% → 64.80%, p = 0.008; val +2.2) and **+9.8** at G=8 (52.80% → 62.60%, p = 1.8e-11; val +8.0), making fixed-subset flat MoE the best configuration in the study (T10, T12). Fixed-nested beats fixed-flat by +3.4 at seed 0 (test, p = 0.01) and *loses* by −5.4 at seed 1 (p = 3.1e-05), so no hierarchy effect survives replication either. **Routing is the cost; nesting only buys granularity tolerance.** Strict sharing of the sub-cell across scale (Variant B) reproduces the old failure on a new axis: −81% parameters, identical MACs/item, **−6.1 points**; and its

control shows why per-slot conditioning is mandatory — remove it and slot outputs become exactly identical (cos = 1.0000, swapping the router's choice changes the stream by 0.0000), i.e. a naive strict fractal has an inert outer router by construction.

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1. What the previous work tested, and what this work tests

The architecture that was measured before (`hmoe.model.FractalNet`) is **one function applied eight times**: a single cell body (`share_weights=True`) or one body per level, unrolled along a single axis — depth — with every application reading and writing the same residual stream at the same width. Its self-similarity is *temporal*. The reported `cos(update_l, update_{l+1}) ≈ 0.85` for the shared cell is not evidence about self-similar organisation; it is a property of a loop: apply one function to a state that function just shaped and consecutive updates point nearly the same way. Nothing in that design can separate "fractal" from "while". What was specified instead is a **nesting**: the outer level routes over experts, and each expert slot contains another complete routed cell — its own intention state, its own router of the same topology, its own pool of sub-experts — which returns a weighted sum of their outputs to the scale that called it. Its self-similarity is in *scale*: the "router choosing among experts" computation appears again one magnification down, inside each thing being routed above, and the base case of the recursion is a plain FFN rather than another invocation of the caller. The two objects differ along four axes at once: the recursion axis (step vs scale), what is shared (one body's parameters vs the shape of the routing computation), arity (a loop application has one state; a nested cell receives a caller's state and returns an update computed by a decision procedure the caller does not see), and what would count as evidence (coherence across steps vs differentiation across scales). A negative result on the first object transfers to the second only if scale-recursion is additionally shown to buy nothing over flat routing at matched parameters, active parameters and MACs. That test had never been run. It is run here: **Condition 3** (hierarchical fractal) against **Condition 1** (flat top-2 MoE), with **Condition 2** (the weight-tied loop) cited from the previous work and not rerun.

2. What was built

`src/hmoe/hier.py`. Two nets, one trunk: `FlatMoENet` is `HierNet` with `nested=False`, so the comparison shares the attention block, the intention update, the router class, the pre-expert LayerNorm and the top-k routine *literally* — a difference in results cannot be attributed to a different trunk.

- **`Router`** — the same module class instantiated at both scales: LayerNorm'd intention → linear → logit-RMS pinned → temperature → non-differentiable load bias. "Same topology at every scale" is enforced by construction, not asserted.
- **`LeafFFN`** — plain SwiGLU; the base case of the recursion. Nothing routes below it.
- **`SubCell`** — a routed cell used *as an expert*: `L_inner=2` sub-levels, `E_inner=4` sub-experts, its own intention state seeded from the caller's intention vector, `k_inner=2` top-2 routing, returning the **delta** it computed. It does not mix tokens across positions: attention exists only at the outer scale, so nesting adds per-token depth/width bookkeeping, not a second attention stack.
- **Top-k with renormalisation, executed as top-k.** Routing is hard top-2 with per-expert token gathering in both training and inference (`route_over`), so the trained function and the executed function coincide — the lesson of FINDINGS §4.9 (soft-mix training: 49.5% token agreement vs 99.5% hard). The router still trains because the renormalised top-2 weights are softmax functions of `z`; unselected experts move via the Switch-style aux term plus the load bias.

- `ln_expert=True` **is enforced by the constructor** (raises otherwise). §5 of the previous report applies to any routed expert, at either scale; there is no supported way to run this work without the norm.

Outer cells are **never** shared across outer depth in any variant — `HierNet` has no code path for it. That is deliberate: it is the design that already failed, and it must stay refutable.

3. Two structural findings established before training

3.1 Strict cross-scale sharing without per-slot conditioning is degenerate

If two expert slots at some level are literally the same module, they compute the same function of the same input, so *which one the caller's router selects carries no information about the computation performed* — only the gate scalars differ. This is a property of the architecture, not of a training run, and it means Variant B ("the sub-cell function is the SAME function at every scale") is only meaningful if slots are conditioned on their identity. Here: a learned **slot embedding** owned by the calling level and added to the sub-cell's intention seed — never shared, even when the body is.

Causal test (`scripts/slot_degeneracy.py`, random init, calc_mix batch, all 8 outer levels):

condition	mean pairwise cos(slot_i(x), slot_j(x))	relative change if the router had picked another slot
Variant B with per-slot conditioning	see T5	see T5
Variant B with slot embeddings **zeroed**	**1.0000**	**0.0000**

With the conditioning removed, the slots are *exactly* interchangeable and swapping the selected slot changes the residual stream by exactly zero: the outer router's discrete choice is inert. This is the failure mode of a naive fractal, found by construction plus a control rather than by observing a bad run.

3.2 Nesting does not change the bill — it can be matched exactly

Tokens are mixed across positions only at the outer scale, so Condition 3 can match Condition 1 on all three required axes simultaneously: with `E_outer·E_inner / (k_outer·k_inner) = E/k = 4`, the total-to-active expert ratio is identical to the flat model's, and one width knob (`e_leaf`) fixes both total parameters and active FFN units per token. `scripts/matched_hier.py` searches this and finds it (T1): all three axes within 0.7%. No axis had to be traded off; if the hypothesis fails, it fails on organisation, not accounting.

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4. Measurement definitions (stated before results, because each one could be gamed)

Cost. `MACs/item = MACs/token × mean(prompt+target tokens)`, sequence length measured from `data/calc_mix/test.meta.jsonl` (n=5,000, mean **44.31**). The analytic cost model counts what the executed code path does: per outer level, attention ($4d^2$), intention update, router(s), and exactly `k_outer` executed leaves — or nested, `k_outer` slots × `L_inner` sub-levels × `k_inner` leaves of width `e_leaf`, **plus the inner routers and inner intention updates**, because those really execute. Nesting pays its rent on this axis; nothing is excluded to flatter Condition 3.

Active parameters per token. Parameters actually touched on a forward pass: attention + intention + router at each outer level, plus the seed and body of each selected slot and the `k_inner` leaves executed at each sub-level. (An early version of this counter multiplied the sub-cell body by `L_inner` *after* already summing over sub-levels and overstated Condition 3's per-token cost by ~60%; fixed, and the fix is visible in `hier.py:active_params_per_token`.)

Settling depth. There is deliberately **no learned halting head** in either condition. The previous work measured that a depth gate free to close from step 0 always closes (executed depth 1.94 with exact match pinned at 12%), and adding one here would confound "does nesting route better?" with "does ACT work?". Settling is therefore a *post-hoc headroom* measurement on a full-depth-trained model, defined two ways and reported separately: - *stream-based*: the last outer level `l` at which the token's top-2 set changed **or** `‖δ_l‖/‖x_l‖ > ε_rel` (ε_rel = 0.10). "The stream stopped moving." - *router-only*: the last outer level at which the token's top-2 **set** changed. "The router stopped deciding." This is the one that tests the task's claim — early settling on easy items — without letting residual-stream decay masquerade as a routing decision.

Router entropy. Per-token mean softmax entropy of each router, in nats, reported against `log E` for that scale, plus the marginal load distribution across the run. Near-uniform at both scales = inert structure (FINDINGS §4.4's "alive but barely selective"). Conditional sub-router entropy given the outer slot chosen is reported per (level, slot): `H(inner | slot) ≪ H(inner)` is hierarchical specialisation; equality means the outer choice does not change what the inner router does.

Specialisation. Selection distributions conditioned on (i) `n_ops` difficulty bucket, (ii) nearest operator to the left of the token (add/sub/mul/div/answer), (iii) digit index within the current operand, (iv) operand magnitude (`max_digits`, 1–3). For (iv) note honestly: calc_mix records no explicit carry field, so operand magnitude is a **proxy** for carry structure, named as a proxy. Conditioning strength is quantified by Jensen-Shannon divergence between bucket distributions (0 = no conditioning at all).

Forced-depth / oracle. Accuracy per difficulty bucket at exactly `k` executed **outer** levels with inner depth always full (`forward_fixed`), cost charged from the model's own cost model at that depth. The oracle then picks the best depth per bucket subject to minimum expected MACs/item for a target accuracy — the same construction as FINDINGS §3, so the two papers' oracle numbers are directly comparable. Both the ceiling (each bucket at its own best depth) and the min-cost targets are reported, together with the chosen depths, because a saving harvested by dumping a hopeless bucket to minimum depth measures hopelessness, not routing (§3 of the previous report).

Update coherence across scales. `cos(δ_outer, δ_inner)` where `δ_outer` is the update a slot returns to the outer stream and `δ_inner` is the *mean per-sub-level* update inside that same slot, compared on the token subset the slot actually served. Plus the direct analogue of the previous work's headline coherence number: `cos(δ_l, δ_{l+1})` across outer depth, and (nested only) `cos` across inner sub-depth within one sub-cell.

5. Conditions and what was matched

id	architecture	status
Condition 1	flat MoE: L=8 levels, E=8 plain-FFN experts, top-2, `ln_expert` on	rerun for this work (task requires the `ln_expert`-on reference)
Condition 1b	granularity control: flat, L=8, E=32 leaves of width 31, top-8	new control (this work)
Condition 2	weight-tied loop: one cell applied 8 times	

id	architecture	status
		cited, not rerun (`FINDINGS_AND_THEORY.md` §4.2/ §5.9, `PAPER_1_matched_cost.md`)
Condition 3A	hierarchical fractal: L_outer=8, E_outer=4 routed cells, L_inner=2, E_inner=4, top-2 at both scales, own parameters per scale	new
Condition 3B	as 3A but ONE sub-cell body shared across every slot and scale, differentiated only by per-slot conditioning (§3.1)	new
Ladder rungs	flat $G \in \{2,4,8,16\}$ (E=8/16/32/64, e=128/62/31/15, top-k=k) vs nested $G \in \{8,16\}$ (`FR_A`, `FR_A_g16`)	new (granularity question)
Long budget	`FR_flat_long`, `FR_A_long`: the headline pair retrained at 18,000 steps	new (budget question)
Inner-routing control	`FR_A_fixed`: nested model with every sub-router replaced by a FIXED random k_inner-subset average — identical params (3,725,088) and MACs/item (60,009,337), content-independent inner choice	new (mechanism question)
Seed replicates	seed 1 reruns of `FR_flat`, `FR_A`, `FR_flat1b`, `FR_flat_g16`, `FR_A_g16`, `FR_A_fixed`, `FR_flat_long`, `FR_A_long`	new
Reciprocal fixed-subset controls	`--outer-route fixed`: flat MoE with its outer top-k softmax replaced by a fixed random k-leaf average (`FR_flat_fixed`, `FR_flat1b_fixed`, and the seed-1 replicate). Mirrors `FR_A_fixed` on the flat side, so "fixed subsets help" can be separated from "nesting helps". Drawn from a private RNG (`torch.Generator().manual_seed(7919 + 31·E + k)`) so that adding the control does not perturb the weight-init RNG stream of pre-existing flat runs	new

Matching deviations, stated exactly (task rule: name what was matched and what was traded). Headline pair `FR_flat` vs `FR_A`: params −0.67%, active/token −0.40%, MACs/item −0.56% — all three matched to better than 1%. Long-budget pair: identical deviations (same architectures, 18,000 steps). Same-granularity pair `FR_flat1b` vs `FR_A`: +1.84% params, +0.98% active/token, +0.80% MACs/item. Fine-granularity pair `FR_flat_g16` vs `FR_A_g16`: **+3.62% params, +1.75% active/token, +1.54% MACs/item** — the nested model is slightly *more* expensive there, because holding leaf width at 15 and active width at 240 while keeping `E_outer=4, L_inner=4, E_inner=4` fixes the number of leaf parameters and leaves the routing machinery as the only slack. The +8.0-point result at that rung (test split; +5.9 val) is therefore obtained with 1.5% more compute per item; the paper reports it that way rather than pretending the rung is exactly matched. That rung is also the most seed-sensitive of the matched-granularity pairs: +8.0 at seed 0, +9.6 at seed 1 (T11). Variant B matches on active params/token and MACs/item exactly (0.00%) and is *deliberately* −81.3% on total parameters — that is the variant being tested.

Condition 1b exists because "8 narrow experts selected per level" is a confound: nesting at `k_outer·L_inner·k_inner = 8` leaves of width 31 buys the same leaf count and the same active width without any hierarchy at all. A flat model can be given exactly the same granularity (E=32, top-8), and then the *only* difference left between it and Condition 3A is the nesting — one router per level versus five routers per token, and a caller that must summarise its sub-decision into a single returned delta. Without 1b, "the fractal works" could never be distinguished from "narrow experts plus more selections per level work". See `scripts/composition_depth.py` (T7) for the arithmetic: 1b and 3A have identical leaves/token/level (8), active width units (248) and sequential FFN compositions per token (64); they differ in routers executed per token (1 vs 5) and in whether the second level of selection is *inside* the thing being routed.

6. Results

Condition 2, cited not rerun. The weight-tied loop that this study uses as its conceptual control is reported in `FINDINGS_AND_THEORY.md` §5.8/§5.9 and `PAPER_1_matched_cost.md`; the numbers used below are quoted from there, with their own denominators (calc_mix, 9,000 steps, best-by-exact on 1,000 val items): `LNE_STRICT` — one cell shared across all 8 levels, 490,146 params — reached **48.4% exact / 48.8% answer**; `LNE_D128_long` — the same backbone with a distinct cell per level, 3.75M params — reached **59.8% / 61.7%**; and the coherence measurement on that family is reported in two places with slightly different values, both far above anything seen here: **≈ 0.85** for the shared cell versus ≈ 0.00 for per-level cells (`FINDINGS_AND_THEORY.md` §5.4/§5.9, from `shared_probe_*.json`), and **0.7345** with `ln_expert` on / **0.4631** with it off in the later re-measurement (`research/results/update_coherence.json`; the same file gives −0.0117 and −0.0046 for per-level cells). Quoting both is deliberate: whichever of the two shared-cell values one takes as canonical, it is an order of magnitude above the 0.09 measured across scales in this study.

(All numbers in this section are emitted by `scripts/hier_tables.py` into `research/results/hier_tables.md`; the tables below are copied from that file. T1 = matched comparison, T2 = stability gate, T3 = routing measurements, T4 = forced-depth/oracle, T4a = teacher-forcing vs rollout, T5 = slot-degeneracy control, T6 = matched-step trajectories, T7 = what nesting changes analytically, T8 = granularity ladder, T9 = seed-0 headline pairs, T10 = inner-routing control, T11 = every headline pair at seed 1, T12 = reciprocal fixed-subset controls on both families.)

T1. Matched-comparison table (all axes named)

Caption: trained on calc_mix, budget = steps in the row, identical optimizer/schedule/seed; accuracy = best-by-exact-match on 1,000 val items; MACs/item = MACs/token x 44.31.

condition	variant	params	active params/ token	MACs/ item	steps	seed	exact	answer	best step
FR_A	A	3,725,088	1,348,352	60,009,337	9,000	0	58.60%	59.80%	9000
FR_A_fixed	A	3,725,088	1,348,352	60,009,337	9,000	0	61.60%	63.00%	9000
FR_A_fixed_s1	A	3,725,088	1,348,352	60,009,337	9,000	1	58.10%	59.00%	9000
FR_A_g16	A	3,696,928	1,341,952	59,714,423	9,000	0	59.70%	61.50%	9000
FR_A_g16_fixed	A	3,696,928	1,341,952	59,714,423	9,000	0	59.60%	60.60%	9000
FR_A_g16_s1	A	3,696,928	1,341,952	59,714,423	9,000	1	56.90%	57.80%	9000
FR_A_long	A	3,725,088	1,348,352	60,009,337	18,000	0	71.20%	72.00%	18000
FR_A_long_s1	A	3,725,088	1,348,352	60,009,337	18,000	1	62.30%	63.60%	18000

condition	variant	params	active params/ token	MACs/ item	steps	seed	exact	answer	best step
FR_A_s1	A	3,725,088	1,348,352	60,009,337	9,000	1	55.60%	56.20%	9000
FR_B	B	700,976	1,348,352	60,009,337	9,000	0	52.50%	52.90%	9000
FR_flat	flat	3,750,048	1,353,728	60,349,623	9,000	0	58.50%	59.30%	9000
FR_flat1b	flat	3,657,888	1,335,296	59,532,938	9,000	0	51.90%	52.60%	9000
FR_flat1b_fixed	flat	3,657,888	1,335,296	59,532,938	9,000	0	59.90%	61.00%	9000
FR_flat1b_s1	flat	3,657,888	1,335,296	59,532,938	9,000	1	51.90%	52.10%	9000
FR_flat_fixed	flat	3,750,048	1,353,728	60,349,623	9,000	0	60.70%	61.70%	9000
FR_flat_fixed_s1	flat	3,750,048	1,353,728	60,349,623	9,000	1	62.50%	63.20%	9000
FR_flat_g16	flat	3,567,776	1,318,912	58,806,995	9,000	0	53.80%	54.90%	9000
FR_flat_g16_s1	flat	3,567,776	1,318,912	58,806,995	9,000	1	48.10%	48.50%	9000
FR_flat_g4	flat	3,653,792	1,331,200	59,351,452	9,000	0	57.00%	57.80%	9000
FR_flat_g4b	flat	3,653,792	1,331,200	59,351,452	9,000	1	55.60%	57.20%	9000
FR_flat_long	flat	3,750,048	1,353,728	60,349,623	18,000	0	64.80%	66.20%	18000
FR_flat_long_s1	flat	3,750,048	1,353,728	60,349,623	18,000	1	63.20%	64.70%	18000
FR_flat_s1	flat	3,750,048	1,353,728	60,349,623	9,000	1	58.80%	59.50%	9000

Deviation from Condition 1 (the matched reference): | variant | d_params | d_active/token | d_MACs/ item | |—|—:|—:|—:| | A | -0.67% | -0.40% | -0.56% | | A | -0.67% | -0.40% | -0.56% | | A | -0.67% | -0.40% | -0.56% | | A | -1.42% | -0.87% | -1.05% | | A | -1.42% | -0.87% | -1.05% | | A | -1.42% | -0.87% | -1.05% | | A | -0.67% | -0.40% | -0.56% | | A | -0.67% | -0.40% | -0.56% | | A | -0.67% | -0.40% | -0.56% | | B | -81.31% | -0.40% | -0.56% | | flat | -2.46% | -1.36% | -1.35% | | flat | -2.46% | -1.36% | -1.35% | | flat | -2.46% | -1.36% | -1.35% | | flat | +0.00% | +0.00% | +0.00% | | flat | +0.00% | +0.00% | +0.00% | | flat | -4.86% | -2.57% | -2.56% | | flat | -4.86% | -2.57% | -2.56% | | flat | -2.57% | -1.66% | -1.65% | | flat | -2.57% | -1.66% | -1.65% | | flat | +0.00% | +0.00% | +0.00% | | flat | +0.00% | +0.00% | +0.00% | | flat | +0.00% | +0.00% | +0.00% | ### T2. Stability gate (init-time RMS, no optimizer step)

Gate from FINDINGS sec 5: refuse any config whose init-time RMS(x) > 1e2. Inner levels are logged separately because nesting adds a second divergence axis.

variant	params	outer RMS by level	max inner RMS	gate	verdict
flat	3,750,048	[0.53, 0.91, 1.24, 1.52, 1.73, 1.91, 2.17, 2.35]	0.0	100	PASS
A	3,725,088	[0.47, 0.82, 1.23, 1.55, 1.86, 2.17, 2.38, 2.57]	2.62	100	PASS
B	700,976		2.28	100	PASS

variant	params	outer RMS by level	max inner RMS	gate	verdict
		[0.45, 0.87, 1.17, 1.44, 1.63, 1.86, 2.07, 2.25]			

Trained-checkpoint growth (RMS(x) at last outer level, from the training log): | run | rms_x_final | NaN steps in log | |—|—:|—:| | FR_A | 7.65 | 0 | | FR_A_fixed | 6.5 | 0 | | FR_A_fixed_s1 | 6.69 | 0 | | FR_A_g16 | 7.18 | 0 | | FR_A_g16_fixed | 6.52 | 0 | | FR_A_g16_s1 | 13.11 | 0 | | FR_A_long | 16.72 | 0 | | FR_A_long_s1 | 13.87 | 0 | | FR_A_s1 | 8.37 | 0 | | FR_B | 4.69 | 0 | | FR_flat | 9.18 | 0 | | FR_flat1b | 5.97 | 0 | | FR_flat1b_fixed | 5.27 | 0 | | FR_flat1b_s1 | 17.54 | 0 | | FR_flat_fixed | 8.59 | 0 | | FR_flat_fixed_s1 | 8.87 | 0 | | FR_flat_g16 | 9.48 | 0 | | FR_flat_g16_s1 | 7.4 | 0 | | FR_flat_g4 | 8.41 | 0 | | FR_flat_g4b | 8.14 | 0 | | FR_flat_long | 10.62 | 0 | | FR_flat_long_s1 | 10.34 | 0 | | FR_flat_s1 | 9.17 | 0 | ### T8. Granularity ladder: nested vs flat at matched active width, varying G

G = sequential leaf compositions per token per outer level (flat: k; nested: k_outer·L_inner·k_inner). Active width units and MACs/item are matched within ~2%; accuracy is best-by-exact on 1,000 val items, 9,000 steps, seed 0 unless noted.

run	family	G	leaves/ token/ level	active width units	params	MACs/ item	exact	answer
FR_flat	flat	2	2	256	3,750,048	60,349,623	58.50%	59.30%
FR_flat_g4	flat	4	4	248	3,653,792	59,351,452	57.00%	57.80%
FR_flat1b	flat	8	8	248	3,657,888	59,532,938	51.90%	52.60%
FR_flat_g16	flat	16	16	240	3,567,776	58,806,995	53.80%	54.90%
FR_A	nested	8	8	248	3,725,088	60,009,337	58.60%	59.80%
FR_A_g16	nested	16	16	240	3,696,928	59,714,423	59.70%	61.50%

T9. Paired per-item significance (McNemar, same test items)

Unpaired binomial SE at n=500 is ±4.4 points, so gaps below ~9 points need a paired test. Both models greedy-generate the SAME 500 calc_mix test items; discordant counts and McNemar p with continuity correction (exact sign test when discordants < 25).

A	B	acc A	acc B	A-only	B-only	p	method
FR_A_fixed_s1	FR_A_s1	60.40%	57.60%	22	8	0.0176	chi2 with continuity correction
FR_A_fixed	FR_A	66.00%	61.00%	32	7	0.000122	chi2 with continuity correction
FR_A_g16_fixed	FR_A_g16	61.20%	63.20%	9	19	0.089	chi2 with continuity correction
FR_flat1b_fixed	FR_A	62.60%	61.00%	22	14	0.243	

A	B	acc A	acc B	A-only	B-only	p	method
							chi2 with continuity correction
FR_flat1b_fixed	FR_A_fixed	62.60%	66.00%	11	28	0.0104	chi2 with continuity correction
FR_flat1b_fixed	FR_flat1b	62.60%	52.80%	50	1	1.8e-11	chi2 with continuity correction
FR_flat1b_s1	FR_A_s1	54.60%	57.60%	9	24	0.0148	chi2 with continuity correction
FR_flat1b	FR_A	52.80%	61.00%	4	45	1.1e-08	chi2 with continuity correction
FR_flat_fixed_s1	FR_A_fixed_s1	65.80%	60.40%	33	6	3.14e-05	chi2 with continuity correction
FR_flat_fixed	FR_A_fixed	64.80%	66.00%	15	21	0.405	chi2 with continuity correction
FR_flat_fixed	FR_flat	64.80%	61.60%	24	8	0.00801	chi2 with continuity correction
FR_flat_g16_s1	FR_A_g16_s1	48.00%	57.60%	7	55	2.39e-09	chi2 with continuity correction
FR_flat_g16	FR_A_g16	55.20%	63.20%	8	48	1.87e-07	chi2 with continuity correction
FR_flat_g4	FR_A	61.60%	61.00%	18	15	0.728	chi2 with continuity correction
FR_flat_long_s1	FR_A_long_s1	66.80%	65.20%	18	10	0.186	chi2 with continuity correction
FR_flat_long	FR_A_long	67.40%	72.20%	13	37	0.00114	chi2 with continuity correction
FR_flat_s1	FR_A_s1	62.20%	57.60%	25	2	2.3e-05	chi2 with continuity correction
FR_flat	FR_A	61.60%	61.00%	17	14	0.719	chi2 with continuity correction

A	B	acc A	acc B	A-only	B-only	p	method
FR_flat	FR_flat1b	61.60%	52.80%	47	3	1.19e-09	chi2 with continuity correction

T10. Content-independent inner routing (routing vs no routing)

`--inner-route fixed` replaces every sub-router with a FIXED random `k_inner`-subset average, identical for all tokens: same executed leaf count, same leaf width, same composition depth, **identical parameter count and MACs/item** as the routed nested model (3,725,088 params / 60,009,337 MACs/item), but the inner choice carries no information about the input. If it matches `FR_A`, the nested win is recursive *width*, not routing.

model	inner routing	params	MACs/item	exact	answer
FR_A	sub-router (content-dependent)	3,725,088	60,009,337	58.60%	59.80%
FR_A_fixed	fixed random subset (content-independent)	3,725,088	60,009,337	61.60%	63.00%

Difference (routed − fixed) = **-3.00 points** on 1,000 val items (unpaired SE ≈ ±3.1). Interpretation: the inner router contributes nothing POSITIVE — replacing it with a fixed random subset IMPROVES accuracy, so the sub-router is actively harmful, not merely inert. The nested win at matched granularity survives with the sub-router removed; see T12 for the reciprocal control on the flat family. Paired test: -3.00 points on val; on test 66.00% (fixed) vs 61.00% (routed), discordant 32/7, McNemar p=0.000122. ### T12. Reciprocal controls: does fixed-subset averaging help the FLAT family too?

`FR_A_fixed` beating `FR_A` cannot be attributed to nesting until the same trick is applied to the flat family, which has the same gate-value and ensembling structure available. `--outer-route fixed` replaces the outer top-k softmax with a fixed random k-leaf average at identical params and MACs/item. If it also lifts flat MoE, the effect is gate values / ensembling rather than hierarchy.

comparison	routed (val)	fixed-subset (val)	delta
flat G=2: routed vs fixed outer subset	58.50%	60.70%	+2.20 pt
flat G=8: routed vs fixed outer subset	51.90%	59.90%	+8.00 pt
nested G=8: routed vs fixed inner subsets	58.60%	61.60%	+3.00 pt
nested G=16: routed vs fixed inner subsets	59.70%	59.60%	-0.10 pt

Paired tests (500 test items, McNemar):

pair	acc A	acc B	A-only	B-only	p
	64.80%	61.60%	24	8	0.00801

pair	acc A	acc B	A-only	B-only	p
FR_flat_fixed vs FR_flat					
FR_flat1b_fixed vs FR_flat1b	62.60%	52.80%	50	1	1.8e-11
FR_A_fixed vs FR_A	66.00%	61.00%	32	7	0.000122
FR_A_g16_fixed vs FR_A_g16	61.20%	63.20%	9	19	0.089
FR_flat1b_fixed vs FR_A	62.60%	61.00%	22	14	0.243
FR_flat1b_fixed vs FR_A_fixed	62.60%	66.00%	11	28	0.0104
FR_flat_fixed_s1 vs FR_A_fixed_s1	65.80%	60.40%	33	6	3.14e-05

Seed-1 replicate of the reciprocal control, and fixed-vs-fixed across families:

comparison	seed 0 routed	seed 0 fixed	seed 1 routed	seed 1 fixed
FR_flat (routed vs fixed-subset)	58.50%	60.70%	58.80%	62.50%
FR_A (routed vs fixed-subset)	58.60%	61.60%	55.60%	58.10%

T11. Seed replicates (seed 1) of the headline comparisons

The G=8 result is a *tie* claim and the fine-granularity result is a *win* claim, so both need replication. Same matched configurations, seed 1, 9,000 steps, best-by-exact on 1,000 val items.

comparison	seed 0 flat	seed 0 nested	seed 1 flat	seed 1 nested
coarse flat G=2 vs nested G=8	58.50%	58.60%	58.80%	55.60%
same granularity G=8	51.90%	58.60%	51.90%	55.60%
fine granularity G=16	53.80%	59.70%	48.10%	56.90%
long budget 18k: coarse flat vs nested	64.80%	71.20%	63.20%	62.30%

T7. What nesting actually changes (analytic, read off the built models)

At matched active width and matched MACs/item, the designs differ in how the same number of FFN units is *arranged* per token. `active width units` is matched by construction; `sequential compositions` is what nesting buys.

condition	params	active/ token	MACs/ item	leaves/ token/ level	active width units	sequential FFN compositions/ token	routers/ token
Condition 1 flat MoE (L=8,E=8,top-2)	3,750,048	1,353,728	60,349,623	2	256	16	1
Condition 1b granularity control (L=8,E=32,e=31,top-8)	3,657,888	1,335,296	59,532,938	8	248	64	1
Condition 3A hierarchical (E_out=4,L_in=2,E_in=4,e_leaf=31)	3,725,088	1,348,352	60,009,337	8	248	64	5
Condition 3B strict cross-scale sharing	700,976	1,348,352	60,009,337	8	248	64	5

T5. Slot-degeneracy control (is the caller's choice informative?)

`scripts/slot_degeneracy.py`: every slot is run on the SAME tokens at each outer level; cos=1 means slots are interchangeable, i.e. the outer router's discrete selection carries no information about the computation.

checkpoint	condition	mean pairwise cos(slot_i(x), slot_j(x))	relative change if a different slot had been selected
FR_B @ random init	share_across_scale=True	0.8144	0.4127
FR_B @ init, slot embeddings ZEROED (control)	share_across_scale=True	1.0000	0.0000
FR_B trained	share_across_scale=True	0.8858	0.2337
FR_A trained	share_across_scale=False	0.0094	1.6133

T3. Routing measurements

FR_A (variant A, n=400 test items, dataset calc_mix)

Settling depth (post-hoc headroom for early exit; eps_rel=0.1):

bucket	mean levels used (stream)	mean (router-only)	frac <=4 (stream)	n tokens
trivial	7.78	7.38	0.0%	16,435
easy	7.66	7.13	0.0%	20,045
medium	7.66	7.36	0.0%	5,527
hard	7.88	7.79	0.0%	3,385

Router entropy per scale: outer H = 0.358 nats (max 1.386); inner H = 0.597 nats (max 1.386)

quantity	value
JSD(outer selection, trivial vs hard)	0.00075
JSD(inner selection, trivial vs hard)	0.00009
JSD(outer selection, 1-digit vs max-digit operands) [carry proxy]	0.00019
JSD(inner selection, 1-digit vs max-digit operands) [carry proxy]	0.00115

Selection distribution by nearest operator (does routing specialise by op type? uniform => no): | scale | op | p(expert_0..k) | H nats | |—|—|—|—:| | outer | add | [0.219, 0.237, 0.221, 0.322] | 1.373 | | outer | answer | [0.357, 0.294, 0.223, 0.126] | 1.323 | | outer | mul | [0.228, 0.217, 0.224, 0.331] | 1.370 | | outer | none | [0.337, 0.234, 0.236, 0.194] | 1.365 | | outer | sub | [0.212, 0.238, 0.219, 0.331] | 1.369 | | inner | add | [0.272, 0.25, 0.23, 0.248] | 1.384 | | inner | answer | [0.242, 0.25, 0.24, 0.268] | 1.385 | | inner | mul | [0.262, 0.252, 0.236, 0.251] | 1.386 | | inner | none | [0.287, 0.24, 0.228, 0.244] | 1.382 | | inner | sub | [0.27, 0.248, 0.235, 0.246] | 1.385 | | cos(update at outer scale, update at inner scale) | 0.9998 | | cos(update_l, update_{l+1}) across OUTER depth | 0.093 | | cos(inner-scale update field at level l, same field at l+1) [the non-tautological cross-scale coherence] | 0.093 | | cos across INNER sub-depth inside one sub-cell | 0.0819 | | frac of tokens whose sub-router kept the same top-2 across sub-levels | 0.187 |

Growth map on the trained checkpoint (RMS(x); outer axis and inner axis logged separately — nesting adds a scale axis along which divergence can occur): | outer level | RMS(x) after level | max inner RMS inside that level | |—:|—:|—:| | 0 | 1.90 | 5.66 | | 1 | 2.68 | 3.94 | | 2 | 2.81 | 4.19 | | 3 | 2.85 | 3.85 | | 4 | 3.11 | 3.28 | | 5 | 3.47 | 3.66 | | 6 | 5.67 | 7.64 | | 7 | 7.93 | 12.33 |

Sub-router entropy conditioned on the outer slot chosen (hierarchical specialisation test; compare with marginal inner H above): | outer level/slot | inner H (nats) | max | |—|—:|—:| | L0S0 | 1.385 | 1.386 | | L0S1 | 1.370 | 1.386 | | L0S2 | 1.376 | 1.386 | | L0S3 | 1.375 | 1.386 | | L1S0 | 1.376 | 1.386 | | L1S1 | 1.367 | 1.386 | | L1S2 | 1.366 | 1.386 | | L1S3 | 1.347 | 1.386 | | L2S0 | 1.364 | 1.386 | | L2S1 | 1.375 | 1.386 | | L2S2 | 1.376 | 1.386 | | L2S3 | 1.373 | 1.386 | | L3S0 | 1.370 | 1.386 | | L3S1 | 1.382 | 1.386 | | L3S2 | 1.384 | 1.386 | | L3S3 | 1.375 | 1.386 | | L4S0 | 1.342 | 1.386 | | L4S1 | 1.375 | 1.386 | | L4S2 | 1.361 | 1.386 | | L4S3 | 1.368 | 1.386 | | L5S0 | 1.383 | 1.386 | | L5S1 | 1.353 | 1.386 | | L5S2 | 1.380 | 1.386 | | L5S3 | 1.375 | 1.386 | | L6S0 | 1.372 | 1.386 | | L6S1 | 1.375 | 1.386 | | L6S2 | 1.352 | 1.386 | | L6S3 | 1.380 | 1.386 | | L7S0 | 1.301 | 1.386 | | L7S1 | 1.372 | 1.386 | | L7S2 | 1.379 | 1.386 | | L7S3 | 1.375 | 1.386 |

FR_A_fixed (variant A, n=400 test items, dataset calc_mix)

Settling depth (post-hoc headroom for early exit; eps_rel=0.1):

bucket	mean levels used (stream)	mean (router-only)	frac <=4 (stream)	n tokens
trivial	7.97	7.74	0.0%	16,435
easy	7.97	7.61	0.0%	20,045
medium	7.97	7.47	0.0%	5,527
hard	8.00	7.49	0.0%	3,385

Router entropy per scale: outer H = 0.488 nats (max 1.386); inner H = 0.490 nats (max 1.386)

quantity	value
JSD(outer selection, trivial vs hard)	0.00026
JSD(inner selection, trivial vs hard)	0.00029
JSD(outer selection, 1-digit vs max-digit operands) [carry proxy]	0.00026
JSD(inner selection, 1-digit vs max-digit operands) [carry proxy]	0.00113

Selection distribution by nearest operator (does routing specialise by op type? uniform => no): | scale | op | p(expert_0..k) | H nats | |—|—|—|—:| | outer | add | [0.313, 0.243, 0.151, 0.293] | 1.352 | | outer | answer | [0.345, 0.324, 0.231, 0.101] | 1.302 | | outer | mul | [0.309, 0.252, 0.148, 0.291] | 1.352 | | outer | none | [0.321, 0.226, 0.153, 0.3] | 1.349 | | outer | sub | [0.313, 0.238, 0.151, 0.298] | 1.352 | | inner | add | [0.235, 0.309, 0.169, 0.287] | 1.362 | | inner | answer | [0.246, 0.282, 0.194, 0.278] | 1.376 | | inner | mul | [0.238, 0.308, 0.168, 0.286] | 1.362 | | inner | none | [0.227, 0.282, 0.18, 0.31] | 1.366 | | inner | sub | [0.235, 0.309, 0.17, 0.287] | 1.362 | | cos(update at outer scale, update at inner scale) | 1.0 | | cos(update_l, update_{l+1}) across OUTER depth | 0.1378 | | cos(inner-scale update field at level l, same field at l+1) [the non-tautological cross-scale coherence] | 0.1378 | | cos across INNER sub-depth inside one sub-cell | 0.151 | | frac of tokens whose sub-router kept the same top-2 across sub-levels | 0.125 |

Growth map on the trained checkpoint (RMS(x); outer axis and inner axis logged separately — nesting adds a scale axis along which divergence can occur): | outer level | RMS(x) after level | max inner RMS inside that level | |—:|—:|—:| | 0 | 0.92 | 1.22 | | 1 | 1.47 | 1.69 | | 2 | 1.86 | 2.13 | | 3 | 2.04 | 2.55 | | 4 | 2.24 | 2.59 | | 5 | 2.60 | 2.89 | | 6 | 3.30 | 4.13 | | 7 | 6.59 | 10.01 |

Sub-router entropy conditioned on the outer slot chosen (hierarchical specialisation test; compare with marginal inner H above): | outer level/slot | inner H (nats) | max | |—|—:|—:| | L0S0 | 1.040 | 1.386 | | L0S1 | 1.040 | 1.386 | | L0S2 | 0.693 | 1.386 | | L0S3 | 0.693 | 1.386 | | L1S0 | 1.040 | 1.386 | | L1S1 | 1.040 | 1.386 | | L1S2 | 1.040 | 1.386 | | L1S3 | 1.040 | 1.386 | | L2S0 | 1.040 | 1.386 | | L2S1 | 1.040 | 1.386 | | L2S2 | 1.040 | 1.386 | | L2S3 | 1.386 | 1.386 | | L3S0 | 1.386 | 1.386 | | L3S1 | 1.040 | 1.386 | | L3S2 | 1.040 | 1.386 | | L3S3 | 1.386 | 1.386 | | L4S0 | 1.040 | 1.386 | | L4S1 | 1.040 | 1.386 | | L4S2 | 1.040 | 1.386 | | L4S3 | 1.386 | 1.386 | | L5S0 | 1.040 | 1.386 | | L5S1 | 1.040 | 1.386 | | L5S2 | 1.040 | 1.386 | | L5S3 | 0.693 | 1.386 | | L6S0 | 1.040 | 1.386 | | L6S1 | 1.040 | 1.386 | | L6S2 | 0.693 | 1.386 | | L6S3 | 1.040 | 1.386 | | L7S0 | 1.040 | 1.386 | | L7S1 | 1.040 | 1.386 | | L7S2 | 1.040 | 1.386 | | L7S3 | 1.040 | 1.386 |

FR_A_fixed_s1 (variant A, n=400 test items, dataset calc_mix)

Settling depth (post-hoc headroom for early exit; eps_rel=0.1):

bucket	mean levels used (stream)	mean (router-only)	frac <=4 (stream)	n tokens
trivial	7.74	7.39	0.0%	16,435
easy	7.85	7.56	0.0%	20,045
medium	7.93	7.68	0.3%	5,527
hard	7.98	7.83	0.0%	3,385

Router entropy per scale: outer H = 0.403 nats (max 1.386); inner H = 0.476 nats (max 1.386)

quantity	value
JSD(outer selection, trivial vs hard)	0.00039
JSD(inner selection, trivial vs hard)	0.00008
JSD(outer selection, 1-digit vs max-digit operands) [carry proxy]	0.00052
JSD(inner selection, 1-digit vs max-digit operands) [carry proxy]	0.00021

Selection distribution by nearest operator (does routing specialise by op type? uniform => no): | scale | op | p(expert_0..k) | H nats | |—|—|—|—:| | outer | add | [0.229, 0.238, 0.193, 0.339] | 1.364 | | outer | answer | [0.374, 0.329, 0.212, 0.085] | 1.272 | | outer | mul | [0.214, 0.243, 0.182, 0.361] | 1.351 | | outer | none | [0.165, 0.288, 0.143, 0.403] | 1.301 | | outer | sub | [0.233, 0.237, 0.194, 0.336] | 1.365 | | inner | add | [0.241, 0.282, 0.203, 0.274] | 1.378 | | inner | answer | [0.242, 0.29, 0.192, 0.276] | 1.374 | | inner | mul | [0.242, 0.278, 0.202, 0.279] | 1.378 | | inner | none | [0.246, 0.263, 0.216, 0.275] | 1.382 | | inner | sub | [0.24, 0.283, 0.203, 0.275] | 1.378 | | cos(update at outer scale, update at inner scale) | 1.0 | | cos(update_l, update_{l+1}) across OUTER depth | 0.1302 | | cos(inner-scale update field at level l, same field at l+1) [the non-tautological cross-scale coherence] | 0.1302 | | cos across INNER sub-depth inside one sub-cell | 0.1365 | | frac of tokens whose sub-router kept the same top-2 across sub-levels | 0.125 |

Growth map on the trained checkpoint (RMS(x); outer axis and inner axis logged separately — nesting adds a scale axis along which divergence can occur): | outer level | RMS(x) after level | max inner RMS inside that level | |—:|—:|—:| | 0 | 1.39 | 2.48 | | 1 | 1.67 | 1.76 | | 2 | 1.90 | 2.25 | | 3 | 2.10 | 2.42 | | 4 | 2.55 | 4.54 | | 5 | 2.79 | 3.47 | | 6 | 3.28 | 4.15 | | 7 | 6.66 | 10.48 |

Sub-router entropy conditioned on the outer slot chosen (hierarchical specialisation test; compare with marginal inner H above): | outer level/slot | inner H (nats) | max | |—|—:|—:| | L0S0 | 1.386 | 1.386 | | L0S1 | 0.693 | 1.386 | | L0S2 | 1.386 | 1.386 | | L0S3 | 1.040 | 1.386 | | L1S0 | 1.040 | 1.386 | | L1S1 | 1.386 | 1.386 | | L1S2 | 1.386 | 1.386 | | L1S3 | 1.040 | 1.386 | | L2S0 | 1.040 | 1.386 | | L2S1 | 1.040 | 1.386 | | L2S2 | 1.040 | 1.386 | | L2S3 | 1.386 | 1.386 | | L3S0 | 1.040 | 1.386 | | L3S1 | 1.040 | 1.386 | | L3S2 | 0.693 | 1.386 | | L3S3 | 1.386 | 1.386 | | L4S0 | 1.040 | 1.386 | | L4S1 | 1.040 | 1.386 | | L4S2 | 1.040 | 1.386 | | L4S3 | 1.040 | 1.386 | | L5S0 | 1.386 | 1.386 | | L5S1 | 0.693 | 1.386 | | L5S2 | 1.386 | 1.386 | | L5S3 | 1.040 | 1.386 | | L6S0 | 1.040 | 1.386 | | L6S1 | 0.693 | 1.386 | | L6S2 | 1.040 | 1.386 | | L6S3 | 1.386 | 1.386 | | L7S0 | 1.040 | 1.386 | | L7S1 | 1.040 | 1.386 | | L7S2 | 1.040 | 1.386 | | L7S3 | 1.040 | 1.386 |

FR_A_g16 (variant A, n=400 test items, dataset calc_mix)

Settling depth (post-hoc headroom for early exit; eps_rel=0.1):

bucket	mean levels used (stream)	mean (router-only)	frac <=4 (stream)	n tokens
trivial	7.72	7.59	0.0%	16,435
easy	7.83	7.67	0.0%	20,045
medium	7.85	7.67	0.0%	5,527
hard	7.99	7.80	0.0%	3,385

Router entropy per scale: outer H = 0.366 nats (max 1.386); inner H = 0.397 nats (max 1.386)

quantity	value
JSD(outer selection, trivial vs hard)	0.00055
JSD(inner selection, trivial vs hard)	0.00004
JSD(outer selection, 1-digit vs max-digit operands) [carry proxy]	0.00052
JSD(inner selection, 1-digit vs max-digit operands) [carry proxy]	0.00035

Selection distribution by nearest operator (does routing specialise by op type? uniform => no): | scale | op | p(expert_0..k) | H nats | |—|—|—|—:| | outer | add | [0.328, 0.191, 0.192, 0.289] | 1.358 | | outer | answer | [0.262, 0.345, 0.252, 0.142] | 1.342 | | outer | mul | [0.338, 0.167, 0.163, 0.332] | 1.327 | | outer | none | [0.327, 0.194, 0.24, 0.24] | 1.368 | | outer | sub | [0.328, 0.196, 0.19, 0.286] | 1.359 | | inner | add | [0.251, 0.247, 0.255, 0.247] | 1.386 | | inner | answer | [0.256, 0.253, 0.238, 0.253] | 1.386 | | inner | mul | [0.252, 0.254, 0.25, 0.245] | 1.386 | | inner | none | [0.256, 0.265, 0.261, 0.218] | 1.383 | | inner | sub | [0.25, 0.248, 0.256, 0.247] | 1.386 | | cos(update at outer scale, update at inner scale) | 1.0 | | cos(update_l, update_{l+1}) across OUTER depth | 0.0954 | | cos(inner-scale update field at level l, same field at l+1) [the non-tautological cross-scale coherence] | 0.0954 | | cos across INNER sub-depth inside one sub-cell | 0.0648 | | frac of tokens whose sub-router kept the same top-2 across sub-levels | 0.128 |

Growth map on the trained checkpoint (RMS(x); outer axis and inner axis logged separately — nesting adds a scale axis along which divergence can occur): | outer level | RMS(x) after level | max inner RMS inside that level | |—:|—:|—:| | 0 | 1.29 | 1.72 | | 1 | 2.65 | 3.60 | | 2 | 3.38 | 5.35 | | 3 | 3.54 | 4.44 | | 4 | 3.92 | 5.10 | | 5 | 4.15 | 4.56 | | 6 | 4.72 | 6.08 | | 7 | 7.26 | 10.11 |

Sub-router entropy conditioned on the outer slot chosen (hierarchical specialisation test; compare with marginal inner H above): | outer level/slot | inner H (nats) | max | |—|—:|—:| | L0S0 | 1.384 | 1.386 | | L0S1 | 1.379 | 1.386 | | L0S2 | 1.382 | 1.386 | | L0S3 | 1.357 | 1.386 | | L1S0 | 1.386 | 1.386 | | L1S1 | 1.382 | 1.386 | | L1S2 | 1.361 | 1.386 | | L1S3 | 1.386 | 1.386 | | L2S0 | 1.382 | 1.386 | | L2S1 | 1.381 | 1.386 | | L2S2 | 1.381 | 1.386 | | L2S3 | 1.358 | 1.386 | | L3S0 | 1.383 | 1.386 | | L3S1 | 1.384 | 1.386 | | L3S2 | 1.371 | 1.386 | | L3S3 | 1.356 | 1.386 | | L4S0 | 1.382 | 1.386 | | L4S1 | 1.381 | 1.386 | | L4S2 | 1.372 | 1.386 | | L4S3 | 1.370 | 1.386 | | L5S0 | 1.367 | 1.386 | | L5S1 | 1.384 | 1.386 | | L5S2 | 1.385 | 1.386 | | L5S3 | 1.382 | 1.386 | | L6S0 | 1.376 | 1.386 | | L6S1 | 1.385 | 1.386 | | L6S2 | 1.376 | 1.386 | | L6S3 | 1.382 | 1.386 | | L7S0 | 1.385 | 1.386 | | L7S1 | 1.381 | 1.386 | | L7S2 | 1.375 | 1.386 | | L7S3 | 1.369 | 1.386 |

FR_A_g16_fixed (variant A, n=400 test items, dataset calc_mix)

Settling depth (post-hoc headroom for early exit; eps_rel=0.1):

bucket	mean levels used (stream)	mean (router-only)	frac <=4 (stream)	n tokens
trivial	7.90	7.68	0.0%	16,435
easy	7.94	7.75	0.0%	20,045
medium	7.97	7.75	0.0%	5,527
hard	8.00	7.74	0.0%	3,385

Router entropy per scale: outer H = 0.215 nats (max 1.386); inner H = 0.359 nats (max 1.386)

quantity	value
JSD(outer selection, trivial vs hard)	0.00090
JSD(inner selection, trivial vs hard)	0.00004
JSD(outer selection, 1-digit vs max-digit operands) [carry proxy]	0.00080
JSD(inner selection, 1-digit vs max-digit operands) [carry proxy]	0.00009

Selection distribution by nearest operator (does routing specialise by op type? uniform => no): | scale | op | p(expert_0..k) | H nats | |—|—|—|—:| | outer | add | [0.249, 0.258, 0.17, 0.323] | 1.362 | | outer | answer | [0.399, 0.275, 0.231, 0.095] | 1.284 | | outer | mul | [0.235, 0.256, 0.172, 0.336] | 1.359 | | outer | none | [0.429, 0.123, 0.183, 0.264] | 1.283 | | outer | sub | [0.248, 0.257, 0.167, 0.328] | 1.359 | | inner | add | [0.257, 0.266, 0.26, 0.217] | 1.383 | | inner | answer | [0.26, 0.28, 0.25, 0.21] | 1.381 | | inner | mul | [0.257, 0.264, 0.264, 0.215] | 1.383 | | inner | none | [0.253, 0.278, 0.27, 0.199] | 1.378 | | inner | sub | [0.257, 0.265, 0.263, 0.215] | 1.383 | | cos(update at outer scale, update at inner scale) | 1.0 | | cos(update_l, update_{l+1}) across OUTER depth | 0.1367 | | cos(inner-scale update field at level l, same field at l+1) [the non-tautological cross-scale coherence] | 0.1367 | | cos across INNER sub-depth inside one sub-cell | 0.1076 | | frac of tokens whose sub-router kept the same top-2 across sub-levels | 0.2187 |

Growth map on the trained checkpoint (RMS(x); outer axis and inner axis logged separately — nesting adds a scale axis along which divergence can occur): | outer level | RMS(x) after level | max inner RMS inside that level | |—:|—:|—:| | 0 | 1.05 | 1.29 | | 1 | 1.35 | 1.47 | | 2 | 1.52 | 1.93 | | 3 | 1.67 | 2.02 | | 4 | 2.88 | 5.93 | | 5 | 3.10 | 3.24 | | 6 | 3.70 | 4.22 | | 7 | 6.60 | 9.36 |

Sub-router entropy conditioned on the outer slot chosen (hierarchical specialisation test; compare with marginal inner H above): | outer level/slot | inner H (nats) | max | |—|—:|—:| | L0S0 | 1.255 | 1.386 | | L0S1 | 1.321 | 1.386 | | L0S2 | 0.974 | 1.386 | | L0S3 | 1.040 | 1.386 | | L1S0 | 1.321 | 1.386 | | L1S1 | 1.255 | 1.386 | | L1S2 | 1.321 | 1.386 | | L1S3 | 1.321 | 1.386 | | L2S0 | 1.386 | 1.386 | | L2S1 | 1.255 | 1.386 | | L2S2 | 1.321 | 1.386 | | L2S3 | 1.321 | 1.386 | | L3S0 | 1.213 | 1.386 | | L3S1 | 1.040 | 1.386 | | L3S2 | 0.974 | 1.386 | | L3S3 | 1.386 | 1.386 | | L4S0 | 1.321 | 1.386 | | L4S1 | 1.082 | 1.386 | | L4S2 | 1.321 | 1.386 | | L4S3 | 0.974 | 1.386 | | L5S0 | 1.255 | 1.386 | | L5S1 | 1.321 | 1.386 | | L5S2 | 1.082 | 1.386 | | L5S3 | 1.040 | 1.386 | | L6S0 | 1.321 | 1.386 | | L6S1 | 1.213 | 1.386 | | L6S2 | 1.255 | 1.386 | | L6S3 | 1.386 | 1.386 | | L7S0 | 1.082 | 1.386 | | L7S1 | 1.213 | 1.386 | | L7S2 | 1.321 | 1.386 | | L7S3 | 1.040 | 1.386 |

FR_A_g16_s1 (variant A, n=400 test items, dataset calc_mix)

Settling depth (post-hoc headroom for early exit; eps_rel=0.1):

bucket	mean levels used (stream)	mean (router-only)	frac <=4 (stream)	n tokens
trivial	7.51	7.28	1.2%	16,435
easy	7.51	7.10	2.2%	20,045
medium	7.76	6.93	1.6%	5,527
hard	7.99	6.83	0.0%	3,385

Router entropy per scale: outer H = 0.406 nats (max 1.386); inner H = 0.413 nats (max 1.386)

quantity	value
JSD(outer selection, trivial vs hard)	0.00016
JSD(inner selection, trivial vs hard)	0.00003
JSD(outer selection, 1-digit vs max-digit operands) [carry proxy]	0.00014
JSD(inner selection, 1-digit vs max-digit operands) [carry proxy]	0.00016

Selection distribution by nearest operator (does routing specialise by op type? uniform => no): | scale | op | p(expert_0..k) | H nats | |—|—|—|—:| | outer | add | [0.357, 0.149, 0.122, 0.372] | 1.275 | | outer | answer | [0.324, 0.359, 0.245, 0.072] | 1.268 | | outer | mul | [0.341, 0.152, 0.124, 0.382] | 1.280 | | outer | none | [0.393, 0.123, 0.083, 0.401] | 1.197 | | outer | sub | [0.357, 0.147, 0.12, 0.376] | 1.272 | | inner | add | [0.265, 0.262, 0.244, 0.229] | 1.385 | | inner | answer | [0.25, 0.256, 0.25, 0.244] | 1.386 | | inner | mul | [0.267, 0.259, 0.246, 0.228] | 1.385 | | inner | none | [0.234, 0.235, 0.278, 0.252] | 1.384 | | inner | sub | [0.263, 0.266, 0.244, 0.227] | 1.384 | | cos(update at outer scale, update at inner scale) | 1.0 | | cos(update_l, update_{l+1}) across OUTER depth | 0.0889 | | cos(inner-scale update field at level l, same field at l+1) [the non-tautological cross-scale coherence] | 0.0889 | | cos across INNER sub-depth inside one sub-cell | 0.0608 | | frac of tokens whose sub-router kept the same top-2 across sub-levels | 0.1361 |

Growth map on the trained checkpoint (RMS(x); outer axis and inner axis logged separately — nesting adds a scale axis along which divergence can occur): | outer level | RMS(x) after level | max inner RMS inside that level | |—:|—:|—:| | 0 | 1.69 | 3.58 | | 1 | 2.10 | 2.58 | | 2 | 2.55 | 3.36 | | 3 | 2.80 | 3.23 | | 4 | 3.05 | 5.06 | | 5 | 3.46 | 4.23 | | 6 | 3.95 | 4.82 | | 7 | 12.57 | 21.42 |

Sub-router entropy conditioned on the outer slot chosen (hierarchical specialisation test; compare with marginal inner H above): | outer level/slot | inner H (nats) | max | |—|—:|—:| | L0S0 | 1.373 | 1.386 | | L0S1 | 1.382 | 1.386 | | L0S2 | 1.382 | 1.386 | | L0S3 | 1.375 | 1.386 | | L1S0 | 1.383 | 1.386 | | L1S1 | 1.385 | 1.386 | | L1S2 | 1.386 | 1.386 | | L1S3 | 1.376 | 1.386 | | L2S0 | 1.354 | 1.386 | | L2S1 | 1.376 | 1.386 | | L2S2 | 1.386 | 1.386 | | L2S3 | 1.380 | 1.386 | | L3S0 | 1.373 | 1.386 | | L3S1 | 1.381 | 1.386 | | L3S2 | 1.386 | 1.386 | | L3S3 | 1.378 | 1.386 | | L4S0 | 1.367 | 1.386 | | L4S1 | 1.381 | 1.386 | | L4S2 | 1.383 | 1.386 | | L4S3 | 1.375 | 1.386 | | L5S0 | 1.369 | 1.386 | | L5S1 | 1.378 | 1.386 | | L5S2 | 1.385 | 1.386 | | L5S3 | 1.372 | 1.386 | | L6S0 | 1.374 | 1.386 | | L6S1 | 1.380 | 1.386 | | L6S2 | 1.383 | 1.386 | | L6S3 | 1.381 | 1.386 | | L7S0 | 1.376 | 1.386 | | L7S1 | 1.375 | 1.386 | | L7S2 | 1.385 | 1.386 | | L7S3 | 1.367 | 1.386 |

FR_A_long (variant A, n=400 test items, dataset calc_mix)

Settling depth (post-hoc headroom for early exit; eps_rel=0.1):

bucket	mean levels used (stream)	mean (router-only)	frac <=4 (stream)	n tokens
trivial	7.64	7.61	0.7%	16,435
easy	7.71	7.65	0.3%	20,045
medium	7.77	7.69	0.1%	5,527
hard	7.95	7.89	0.0%	3,385

Router entropy per scale: outer H = 0.332 nats (max 1.386); inner H = 0.474 nats (max 1.386)

quantity	value
JSD(outer selection, trivial vs hard)	0.00061
JSD(inner selection, trivial vs hard)	0.00012
JSD(outer selection, 1-digit vs max-digit operands) [carry proxy]	0.00047
JSD(inner selection, 1-digit vs max-digit operands) [carry proxy]	0.00033

Selection distribution by nearest operator (does routing specialise by op type? uniform => no): | scale | op | p(expert_0..k) | H nats | |—|—|—|—:| | outer | add | [0.176, 0.263, 0.221, 0.341] | 1.357 | | outer | answer | [0.31, 0.294, 0.248, 0.148] | 1.352 | | outer | mul | [0.181, 0.261, 0.207, 0.351] | 1.353 | | outer | none | [0.264, 0.208, 0.306, 0.222] | 1.374 | | outer | sub | [0.17, 0.264, 0.218, 0.348] | 1.352 | | inner | add | [0.274, 0.247, 0.219, 0.259] | 1.383 | | inner | answer | [0.248, 0.251, 0.25, 0.25] | 1.386 | | inner | mul | [0.278, 0.251, 0.217, 0.253] | 1.383 | | inner | none | [0.313, 0.246, 0.167, 0.274] | 1.362 | | inner | sub | [0.273, 0.246, 0.222, 0.259] | 1.383 | | cos(update at outer scale, update at inner scale) | 0.9999 | | cos(update_l, update_{l+1}) across OUTER depth | 0.0992 | | cos(inner-scale update field at level l, same field at l+1) [the non-tautological cross-scale coherence] | 0.0992 | | cos across INNER sub-depth inside one sub-cell | 0.0881 | | frac of tokens whose sub-router kept the same top-2 across sub-levels | 0.1987 |

Growth map on the trained checkpoint (RMS(x); outer axis and inner axis logged separately — nesting adds a scale axis along which divergence can occur): | outer level | RMS(x) after level | max inner RMS inside that level | |—:|—:|—:| | 0 | 4.00 | 15.93 | | 1 | 4.66 | 7.89 | | 2 | 5.18 | 7.47 | | 3 | 5.50 | 9.70 | | 4 | 6.06 | 8.17 | | 5 | 7.25 | 12.53 | | 6 | 8.67 | 10.27 | | 7 | 18.09 | 51.80 |

Sub-router entropy conditioned on the outer slot chosen (hierarchical specialisation test; compare with marginal inner H above): | outer level/slot | inner H (nats) | max | |—|—:|—:| | L0S0 | 1.385 | 1.386 | | L0S1 | 1.355 | 1.386 | | L0S2 | 1.378 | 1.386 | | L0S3 | 1.378 | 1.386 | | L1S0 | 1.370 | 1.386 | | L1S1 | 1.376 | 1.386 | | L1S2 | 1.376 | 1.386 | | L1S3 | 1.350 | 1.386 | | L2S0 | 1.357 | 1.386 | | L2S1 | 1.377 | 1.386 | | L2S2 | 1.370 | 1.386 | | L2S3 | 1.386 | 1.386 | | L3S0 | 1.380 | 1.386 | | L3S1 | 1.378 | 1.386 | | L3S2 | 1.382 | 1.386 | | L3S3 | 1.372 | 1.386 | | L4S0 | 1.347 | 1.386 | | L4S1 | 1.348 | 1.386 | | L4S2 | 1.374 | 1.386 | | L4S3 | 1.361 | 1.386 | | L5S0 | 1.376 | 1.386 | | L5S1 | 1.360 | 1.386 | | L5S2 | 1.383 | 1.386 | | L5S3 | 1.379 | 1.386 | | L6S0 | 1.384 | 1.386 | | L6S1 | 1.381 | 1.386 | | L6S2 | 1.372 | 1.386 | | L6S3 | 1.362 | 1.386 | | L7S0 | 1.323 | 1.386 | | L7S1 | 1.383 | 1.386 | | L7S2 | 1.380 | 1.386 | | L7S3 | 1.378 | 1.386 |

FR_A_s1 (variant A, n=400 test items, dataset calc_mix)

Settling depth (post-hoc headroom for early exit; eps_rel=0.1):

bucket	mean levels used (stream)	mean (router-only)	frac <=4 (stream)	n tokens
trivial	7.74	7.70	0.0%	16,435
easy	7.80	7.72	0.0%	20,045
medium	7.87	7.78	0.0%	5,527
hard	7.99	7.90	0.0%	3,385

Router entropy per scale: outer H = 0.514 nats (max 1.386); inner H = 0.514 nats (max 1.386)

quantity	value
JSD(outer selection, trivial vs hard)	0.00024
JSD(inner selection, trivial vs hard)	0.00009
JSD(outer selection, 1-digit vs max-digit operands) [carry proxy]	0.00034
JSD(inner selection, 1-digit vs max-digit operands) [carry proxy]	0.00068

Selection distribution by nearest operator (does routing specialise by op type? uniform => no): | scale | op | p(expert_0..k) | H nats | |—|—|—|—:| | outer | add | [0.275, 0.201, 0.192, 0.332] | 1.360 | | outer | answer | [0.346, 0.335, 0.226, 0.094] | 1.292 | | outer | mul | [0.222, 0.22, 0.203, 0.355] | 1.359 | | outer | none | [0.244, 0.325, 0.188, 0.244] | 1.367 | | outer | sub | [0.28, 0.188, 0.192, 0.339] | 1.355 | | inner | add | [0.27, 0.238, 0.245, 0.247] | 1.385 | | inner | answer | [0.239, 0.254, 0.254, 0.253] | 1.386 | | inner | mul | [0.274, 0.242, 0.241, 0.243] | 1.385 | | inner | none | [0.252, 0.183, 0.297, 0.269] | 1.372 | | inner | sub | [0.276, 0.239, 0.243, 0.243] | 1.385 | | cos(update at outer scale, update at inner scale) | 0.9998 | | cos(update_l, update_{l+1}) across OUTER depth | 0.0996 | | cos(inner-scale update field at level l, same field at l+1) [the non-tautological cross-scale coherence] | 0.0997 | | cos across INNER sub-depth inside one sub-cell | 0.0915 | | frac of tokens whose sub-router kept the same top-2 across sub-levels | 0.1829 |

Growth map on the trained checkpoint (RMS(x); outer axis and inner axis logged separately — nesting adds a scale axis along which divergence can occur): | outer level | RMS(x) after level | max inner RMS inside that level | |—:|—:|—:| | 0 | 1.64 | 4.05 | | 1 | 2.99 | 5.87 | | 2 | 3.29 | 3.38 | | 3 | 3.43 | 4.15 | | 4 | 4.82 | 7.86 | | 5 | 5.07 | 5.92 | | 6 | 5.90 | 7.69 | | 7 | 8.54 | 11.17 |

Sub-router entropy conditioned on the outer slot chosen (hierarchical specialisation test; compare with marginal inner H above): | outer level/slot | inner H (nats) | max | |—|—:|—:| | L0S0 | 1.371 | 1.386 | | L0S1 | 1.359 | 1.386 | | L0S2 | 1.385 | 1.386 | | L0S3 | 1.363 | 1.386 | | L1S0 | 1.370 | 1.386 | | L1S1 | 1.361 | 1.386 | | L1S2 | 1.384 | 1.386 | | L1S3 | 1.378 | 1.386 | | L2S0 | 1.383 | 1.386 | | L2S1 | 1.377 | 1.386 | | L2S2 | 1.337 | 1.386 | | L2S3 | 1.365 | 1.386 | | L3S0 | 1.369 | 1.386 | | L3S1 | 1.350 | 1.386 | | L3S2 | 1.369 | 1.386 | | L3S3 | 1.375 | 1.386 | | L4S0 | 1.363 | 1.386 | | L4S1 | 1.335 | 1.386 | | L4S2 | 1.340 | 1.386 | | L4S3 | 1.369 | 1.386 | | L5S0 | 1.361 | 1.386 | | L5S1 | 1.354 | 1.386 | | L5S2 | 1.379 | 1.386 | | L5S3 | 1.373 | 1.386 | | L6S0 | 1.373 | 1.386 | | L6S1 | 1.358 | 1.386 | | L6S2 | 1.379 | 1.386 | | L6S3 | 1.363 | 1.386 | | L7S0 | 1.356 | 1.386 | | L7S1 | 1.381 | 1.386 | | L7S2 | 1.381 | 1.386 | | L7S3 | 1.383 | 1.386 |

FR_B (variant B, n=400 test items, dataset calc_mix)

Settling depth (post-hoc headroom for early exit; eps_rel=0.1):

bucket	mean levels used (stream)	mean (router-only)	frac <=4 (stream)	n tokens
trivial	7.57	7.53	0.2%	16,435
easy	7.35	7.22	0.3%	20,045
medium	7.41	7.14	0.2%	5,527
hard	7.63	7.25	1.6%	3,385

Router entropy per scale: outer H = 1.153 nats (max 1.386); inner H = 1.309 nats (max 1.386)

quantity	value
JSD(outer selection, trivial vs hard)	0.00007
JSD(inner selection, trivial vs hard)	0.00677
JSD(outer selection, 1-digit vs max-digit operands) [carry proxy]	0.00007
JSD(inner selection, 1-digit vs max-digit operands) [carry proxy]	0.01636

Selection distribution by nearest operator (does routing specialise by op type? uniform => no): | scale | op | p(expert_0..k) | H nats | |—|—|—|—:| | outer | add | [0.325, 0.153, 0.139, 0.384] | 1.294 | | outer | answer | [0.208, 0.331, 0.319, 0.141] | 1.334 | | outer | mul | [0.303, 0.152, 0.126, 0.419] | 1.273 | | outer | none | [0.206, 0.175, 0.181, 0.438] | 1.302 | | outer | sub | [0.336, 0.15, 0.143, 0.372] | 1.296 | | inner | add | [0.292, 0.317, 0.185, 0.206] | 1.361 | | inner | answer | [0.261, 0.245, 0.188, 0.306] | 1.372 | | inner | mul | [0.272, 0.274, 0.154, 0.3] | 1.358 | | inner | none | [0.379, 0.347, 0.199, 0.076] | 1.251 | | inner | sub | [0.291, 0.328, 0.18, 0.201] | 1.356 | | cos(update at outer scale, update at inner scale) | 1.0 | | cos(update_l, update_{l+1}) across OUTER depth | 0.5539 | | cos(inner-scale update field at level l, same field at l+1) [the non-tautological cross-scale coherence] | 0.5539 | | cos across INNER sub-depth inside one sub-cell | 0.0736 | | frac of tokens whose sub-router kept the same top-2 across sub-levels | 0.077 |

Growth map on the trained checkpoint (RMS(x); outer axis and inner axis logged separately — nesting adds a scale axis along which divergence can occur): | outer level | RMS(x) after level | max inner RMS inside that level | |—:|—:|—:| | 0 | 0.98 | 1.12 | | 1 | 1.43 | 1.54 | | 2 | 1.83 | 2.22 | | 3 | 2.09 | 2.26 | | 4 | 2.32 | 2.66 | | 5 | 2.60 | 3.26 | | 6 | 3.12 | 3.93 | | 7 | 4.88 | 6.27 |

Sub-router entropy conditioned on the outer slot chosen (hierarchical specialisation test; compare with marginal inner H above): | outer level/slot | inner H (nats) | max | |—|—:|—:| | L0S0 | 1.349 | 1.386 | | L0S1 | 1.377 | 1.386 | | L0S2 | 1.368 | 1.386 | | L0S3 | 1.379 | 1.386 | | L1S0 | 1.373 | 1.386 | | L1S1 | 1.354 | 1.386 | | L1S2 | 1.372 | 1.386 | | L1S3 | 1.367 | 1.386 | | L2S0 | 1.373 | 1.386 | | L2S1 | 1.363 | 1.386 | | L2S2 | 1.324 | 1.386 | | L2S3 | 1.372 | 1.386 | | L3S0 | 1.374 | 1.386 | | L3S1 | 1.345 | 1.386 | | L3S2 | 1.367 | 1.386 | | L3S3 | 1.374 | 1.386 | | L4S0 | 1.377 | 1.386 | | L4S1 | 1.345 | 1.386 | | L4S2 | 1.371 | 1.386 | | L4S3 | 1.377 | 1.386 | | L5S0 | 1.374 | 1.386 | | L5S1 | 1.374 | 1.386 | | L5S2 | 1.371 | 1.386 | | L5S3 | 1.374 | 1.386 | | L6S0 | 1.373 | 1.386 | | L6S1 | 1.377 | 1.386 | | L6S2 | 1.374 | 1.386 | | L6S3 | 1.379 | 1.386 | | L7S0 | 1.380 | 1.386 | | L7S1 | 1.377 | 1.386 | | L7S2 | 1.365 | 1.386 | | L7S3 | 1.374 | 1.386 |

FR_flat (variant flat, n=400 test items, dataset calc_mix)

Settling depth (post-hoc headroom for early exit; eps_rel=0.1):

bucket	mean levels used (stream)	mean (router-only)	frac <=4 (stream)	n tokens
trivial	7.94	7.90	0.0%	16,435
easy	7.91	7.85	0.0%	20,045
medium	7.98	7.89	0.0%	5,527
hard	8.00	7.98	0.0%	3,385

Router entropy per scale: outer H = 1.375 nats (max 2.079); inner H = - nats

quantity	value
JSD(outer selection, trivial vs hard)	0.00113
JSD(inner selection, trivial vs hard)	-
JSD(outer selection, 1-digit vs max-digit operands) [carry proxy]	0.00462
JSD(inner selection, 1-digit vs max-digit operands) [carry proxy]	-

Selection distribution by nearest operator (does routing specialise by op type? uniform => no): | scale | op | p(expert_0..k) | H nats | |—|—|—|—:| | outer | add | [0.205, 0.125, 0.156, 0.073, 0.091, 0.114, 0.113, 0.122] | 2.036 | | outer | answer | [0.138, 0.132, 0.109, 0.16, 0.138, 0.147, 0.106, 0.07] | 2.055 | | outer | mul | [0.174, 0.118, 0.147, 0.082, 0.088, 0.124, 0.103, 0.164] | 2.047 | | outer | none | [0.252, 0.141, 0.137, 0.04, 0.065, 0.109, 0.091, 0.165] | 1.959 | | outer | sub | [0.208, 0.131, 0.157, 0.077, 0.092, 0.106, 0.11, 0.12] | 2.034 | | cos(update at outer scale, update at inner scale) | - | | cos(update_l, update_{l+1}) across OUTER depth | 0.1331 | | cos(inner-scale update field at level l, same field at l+1) [the non-tautological cross-scale coherence] | - | | cos across INNER sub-depth inside one sub-cell | - | | frac of tokens whose sub-router kept the same top-2 across sub-levels | - |

Growth map on the trained checkpoint (RMS(x); outer axis and inner axis logged separately — nesting adds a scale axis along which divergence can occur): | outer level | RMS(x) after level | max inner RMS inside that level | |—:|—:|—:| | 0 | 1.28 | - | | 1 | 1.63 | - | | 2 | 2.12 | - | | 3 | 2.56 | - | | 4 | 2.87 | - | | 5 | 3.43 | - | | 6 | 4.41 | - | | 7 | 9.42 | - |

FR_flat1b (variant flat, n=400 test items, dataset calc_mix)

Settling depth (post-hoc headroom for early exit; eps_rel=0.1):

bucket	mean levels used (stream)	mean (router-only)	frac <=4 (stream)	n tokens
trivial	8.00	8.00	0.0%	16,435
easy	8.00	8.00	0.0%	20,045
medium	8.00	8.00	0.0%	5,527
hard	8.00	8.00	0.0%	3,385

Router entropy per scale: outer H = 1.810 nats (max 3.466); inner H = - nats

quantity	value
JSD(outer selection, trivial vs hard)	0.00169
JSD(inner selection, trivial vs hard)	-
JSD(outer selection, 1-digit vs max-digit operands) [carry proxy]	0.00153
JSD(inner selection, 1-digit vs max-digit operands) [carry proxy]	-

Selection distribution by nearest operator (does routing specialise by op type? uniform => no): | scale | op | p(expert_0..k) | H nats | |—|—|—|—:| | outer | add | [0.022, 0.034, 0.04, 0.039, 0.032, 0.028, 0.033,

0.033, 0.028, 0.035, 0.019, 0.034, 0.039, 0.032, 0.041, 0.022, 0.02, 0.023, 0.019, 0.029, 0.021, 0.031, 0.032, 0.026, 0.038, 0.029, 0.032, 0.03, 0.043, 0.039, 0.039, 0.036] | 3.441 | | outer | answer | [0.044, 0.038, 0.026, 0.025, 0.033, 0.043, 0.029, 0.032, 0.038, 0.036, 0.044, 0.031, 0.036, 0.029, 0.025, 0.044, 0.032, 0.043, 0.043, 0.025, 0.045, 0.038, 0.04, 0.03, 0.036, 0.037, 0.015, 0.02, 0.026, 0.007, 0.007, 0.003] | 3.386 | | outer | mul | [0.022, 0.026, 0.035, 0.042, 0.027, 0.033, 0.04, 0.028, 0.029, 0.026, 0.015, 0.041, 0.035, 0.036, 0.036, 0.017, 0.019, 0.018, 0.017, 0.042, 0.022, 0.031, 0.031, 0.025, 0.042, 0.027, 0.027, 0.033, 0.044, 0.04, 0.046, 0.046] | 3.423 | | outer | none | [0.014, 0.042, 0.029, 0.009, 0.032, 0.022, 0.041, 0.025, 0.02, 0.043, 0.027, 0.018, 0.019, 0.025, 0.024, 0.031, 0.047, 0.018, 0.029, 0.025, 0.021, 0.043, 0.022, 0.046, 0.017, 0.032, 0.033, 0.039, 0.049, 0.072, 0.054, 0.032] | 3.379 | | outer | sub | [0.023, 0.035, 0.038, 0.039, 0.03, 0.028, 0.031, 0.034, 0.028, 0.035, 0.019, 0.034, 0.04, 0.033, 0.042, 0.023, 0.019, 0.024, 0.018, 0.032, 0.021, 0.03, 0.032, 0.024, 0.038, 0.03, 0.032, 0.031, 0.044, 0.038, 0.039, 0.037] | 3.440 | | cos(update at outer scale, update at inner scale) | - | | cos(update_l, update_{l+1}) across OUTER depth | 0.0903 | | cos(inner-scale update field at level l, same field at l+1) [the non-tautological cross-scale coherence] | - | | cos across INNER sub-depth inside one sub-cell | - | | frac of tokens whose sub-router kept the same top-2 across sub-levels | - |

Growth map on the trained checkpoint (RMS(x); outer axis and inner axis logged separately — nesting adds a scale axis along which divergence can occur): | outer level | RMS(x) after level | max inner RMS inside that level | |—:|—:|—:| | 0 | 1.76 | - | | 1 | 2.31 | - | | 2 | 2.70 | - | | 3 | 2.76 | - | | 4 | 3.27 | - | | 5 | 3.36 | - | | 6 | 4.40 | - | | 7 | 6.08 | - |

FR_flat1b_fixed (variant flat, n=400 test items, dataset calc_mix)

Settling depth (post-hoc headroom for early exit; eps_rel=0.1):

bucket	mean levels used (stream)	mean (router-only)	frac <=4 (stream)	n tokens
trivial	8.00	1.00	0.0%	16,435
easy	8.00	1.00	0.0%	20,045
medium	8.00	1.00	0.0%	5,527
hard	8.00	1.00	0.0%	3,385

Router entropy per scale: outer H = 2.755 nats (max 3.466); inner H = - nats

quantity	value
JSD(outer selection, trivial vs hard)	0.00000
JSD(inner selection, trivial vs hard)	-
JSD(outer selection, 1-digit vs max-digit operands) [carry proxy]	0.00000
JSD(inner selection, 1-digit vs max-digit operands) [carry proxy]	-

Selection distribution by nearest operator (does routing specialise by op type? uniform => no): | scale | op | p(expert_0..k) | H nats | |—|—|—|—:| | outer | add | [0.0, 0.0, 0.0, 0.125, 0.0, 0.0, 0.0, 0.0, 0.125, 0.0, 0.125, 0.0, 0.125, 0.0, 0.0, 0.0, 0.0, 0.125, 0.0, 0.0, 0.0, 0.125, 0.0, 0.0, 0.0, 0.0, 0.125, 0.125, 0.0, 0.0, 0.0, 0.0] | 2.079 | | outer | answer | [0.0, 0.0, 0.0, 0.125, 0.0, 0.0, 0.0, 0.0, 0.125, 0.0, 0.125, 0.0, 0.125, 0.0, 0.0, 0.0, 0.0, 0.125, 0.0, 0.0, 0.0, 0.125, 0.0, 0.0, 0.0, 0.0, 0.125, 0.125, 0.0, 0.0, 0.0, 0.0] | 2.079 | | outer | mul | [0.0, 0.0, 0.0, 0.125, 0.0, 0.0, 0.0, 0.0, 0.125, 0.0, 0.125, 0.0, 0.125, 0.0, 0.0, 0.0, 0.0, 0.125, 0.0, 0.0, 0.0, 0.125, 0.0, 0.0, 0.0, 0.0, 0.125, 0.125, 0.0, 0.0, 0.0, 0.0] | 2.079 | | outer | none | [0.0, 0.0, 0.0, 0.125, 0.0, 0.0, 0.0, 0.0, 0.125, 0.0,

0.125, 0.0, 0.125, 0.0, 0.0, 0.0, 0.0, 0.125, 0.0, 0.0, 0.0, 0.125, 0.0, 0.0, 0.0, 0.0, 0.125, 0.125, 0.0, 0.0, 0.0, 0.0] | 2.079 | | outer | sub | [0.0, 0.0, 0.0, 0.125, 0.0, 0.0, 0.0, 0.0, 0.125, 0.0, 0.125, 0.0, 0.125, 0.0, 0.0, 0.0, 0.0, 0.125, 0.0, 0.0, 0.0, 0.125, 0.0, 0.0, 0.0, 0.0, 0.125, 0.125, 0.0, 0.0, 0.0, 0.0] | 2.079 | | cos(update at outer scale, update at inner scale) | - | | cos(update_l, update_{l+1}) across OUTER depth | 0.2532 | | cos(inner-scale update field at level l, same field at l+1) [the non-tautological cross-scale coherence] | - | | cos across INNER sub-depth inside one sub-cell | - | | frac of tokens whose sub-router kept the same top-2 across sub-levels | - |

Growth map on the trained checkpoint (RMS(x); outer axis and inner axis logged separately — nesting adds a scale axis along which divergence can occur): | outer level | RMS(x) after level | max inner RMS inside that level | |—:|—:|—:| | 0 | 0.85 | - | | 1 | 1.22 | - | | 2 | 1.67 | - | | 3 | 1.83 | - | | 4 | 1.72 | - | | 5 | 1.83 | - | | 6 | 2.11 | - | | 7 | 5.30 | - |

FR_flat1b_s1 (variant flat, n=400 test items, dataset calc_mix)

Settling depth (post-hoc headroom for early exit; eps_rel=0.1):

bucket	mean levels used (stream)	mean (router-only)	frac <=4 (stream)	n tokens
trivial	8.00	8.00	0.0%	16,435
easy	8.00	8.00	0.0%	20,045
medium	8.00	8.00	0.0%	5,527
hard	8.00	8.00	0.0%	3,385

Router entropy per scale: outer H = 2.513 nats (max 3.466); inner H = - nats

quantity	value
JSD(outer selection, trivial vs hard)	0.00110
JSD(inner selection, trivial vs hard)	-
JSD(outer selection, 1-digit vs max-digit operands) [carry proxy]	0.00097
JSD(inner selection, 1-digit vs max-digit operands) [carry proxy]	-

Selection distribution by nearest operator (does routing specialise by op type? uniform => no): | scale | op | p(expert_0..k) | H nats | |—|—|—|—:| | outer | add | [0.034, 0.026, 0.031, 0.034, 0.028, 0.036, 0.037, 0.031, 0.036, 0.037, 0.029, 0.034, 0.033, 0.026, 0.04, 0.025, 0.027, 0.022, 0.025, 0.032, 0.037, 0.024, 0.033, 0.023, 0.03, 0.031, 0.03, 0.034, 0.03, 0.042, 0.031, 0.03] | 3.454 | | outer | answer | [0.029, 0.032, 0.036, 0.023, 0.027, 0.053, 0.039, 0.035, 0.025, 0.034, 0.037, 0.043, 0.027, 0.035, 0.038, 0.034, 0.034, 0.033, 0.051, 0.025, 0.032, 0.033, 0.042, 0.034, 0.033, 0.05, 0.033, 0.024, 0.013, 0.008, 0.004, 0.005] | 3.381 | | outer | mul | [0.033, 0.031, 0.026, 0.037, 0.031, 0.027, 0.032, 0.024, 0.034, 0.032, 0.026, 0.037, 0.034, 0.025, 0.034, 0.021, 0.03, 0.02, 0.029, 0.035, 0.038, 0.027, 0.038, 0.02, 0.031, 0.039, 0.028, 0.041, 0.032, 0.037, 0.029, 0.042] | 3.448 | | outer | none | [0.047, 0.029, 0.0, 0.023, 0.058, 0.03, 0.019, 0.017, 0.03, 0.018, 0.055, 0.027, 0.068, 0.033, 0.007, 0.036, 0.03, 0.035, 0.029, 0.021, 0.015, 0.065, 0.017, 0.031, 0.055, 0.021, 0.021, 0.045, 0.038, 0.03, 0.016, 0.034] | 3.326 | | outer | sub | [0.033, 0.026, 0.033, 0.036, 0.026, 0.036, 0.037, 0.032, 0.036, 0.038, 0.029, 0.032, 0.03, 0.025, 0.042, 0.024, 0.028, 0.023, 0.024, 0.033, 0.04, 0.027, 0.035, 0.024, 0.029, 0.031, 0.031, 0.032, 0.03, 0.04, 0.03, 0.031] | 3.453 | | cos(update at outer scale, update at inner scale) | - | | cos(update_l, update_{l+1}) across OUTER depth | 0.1013 | | cos(inner-scale update field at level l, same field at l+1)

[the non-tautological cross-scale coherence] | - | | cos across INNER sub-depth inside one sub-cell | - | | frac of tokens whose sub-router kept the same top-2 across sub-levels | - |

Growth map on the trained checkpoint (RMS(x); outer axis and inner axis logged separately — nesting adds a scale axis along which divergence can occur): | outer level | RMS(x) after level | max inner RMS inside that level | |—:|—:|—:| | 0 | 1.11 | - | | 1 | 1.39 | - | | 2 | 1.85 | - | | 3 | 2.11 | - | | 4 | 2.31 | - | | 5 | 2.86 | - | | 6 | 3.26 | - | | 7 | 16.57 | - |

FR_flat_fixed (variant flat, n=400 test items, dataset calc_mix)

Settling depth (post-hoc headroom for early exit; eps_rel=0.1):

bucket	mean levels used (stream)	mean (router-only)	frac <=4 (stream)	n tokens
trivial	8.00	1.00	0.0%	16,435
easy	8.00	1.00	0.0%	20,045
medium	8.00	1.00	0.0%	5,527
hard	8.00	1.00	0.0%	3,385

Router entropy per scale: outer H = 1.734 nats (max 2.079); inner H = - nats

quantity	value
JSD(outer selection, trivial vs hard)	0.00000
JSD(inner selection, trivial vs hard)	-
JSD(outer selection, 1-digit vs max-digit operands) [carry proxy]	0.00000
JSD(inner selection, 1-digit vs max-digit operands) [carry proxy]	-

Selection distribution by nearest operator (does routing specialise by op type? uniform => no): | scale | op | p(expert_0..k) | H nats | |—|—|—|—:| | outer | add | [0.0, 0.0, 0.5, 0.5, 0.0, 0.0, 0.0, 0.0] | 0.693 | | outer | answer | [0.0, 0.0, 0.5, 0.5, 0.0, 0.0, 0.0, 0.0] | 0.693 | | outer | mul | [0.0, 0.0, 0.5, 0.5, 0.0, 0.0, 0.0, 0.0] | 0.693 | | outer | none | [0.0, 0.0, 0.5, 0.5, 0.0, 0.0, 0.0, 0.0] | 0.693 | | outer | sub | [0.0, 0.0, 0.5, 0.5, 0.0, 0.0, 0.0, 0.0] | 0.693 | | cos(update at outer scale, update at inner scale) | - | | cos(update_l, update_{l+1}) across OUTER depth | 0.1926 | | cos(inner-scale update field at level l, same field at l+1) [the non-tautological cross-scale coherence] | - | | cos across INNER sub-depth inside one sub-cell | - | | frac of tokens whose sub-router kept the same top-2 across sub-levels | - |

Growth map on the trained checkpoint (RMS(x); outer axis and inner axis logged separately — nesting adds a scale axis along which divergence can occur): | outer level | RMS(x) after level | max inner RMS inside that level | |—:|—:|—:| | 0 | 1.07 | - | | 1 | 1.61 | - | | 2 | 1.86 | - | | 3 | 2.01 | - | | 4 | 2.30 | - | | 5 | 2.81 | - | | 6 | 3.30 | - | | 7 | 8.62 | - |

FR_flat_fixed_s1 (variant flat, n=400 test items, dataset calc_mix)

Settling depth (post-hoc headroom for early exit; eps_rel=0.1):

bucket	mean levels used (stream)	mean (router-only)	frac <=4 (stream)	n tokens
trivial	8.00	1.00	0.0%	16,435
easy	8.00	1.00	0.0%	20,045
medium	8.00	1.00	0.0%	5,527
hard	8.00	1.00	0.0%	3,385

Router entropy per scale: outer H = 1.587 nats (max 2.079); inner H = - nats

quantity	value
JSD(outer selection, trivial vs hard)	0.00000
JSD(inner selection, trivial vs hard)	-
JSD(outer selection, 1-digit vs max-digit operands) [carry proxy]	0.00000
JSD(inner selection, 1-digit vs max-digit operands) [carry proxy]	-

Selection distribution by nearest operator (does routing specialise by op type? uniform => no): | scale | op | p(expert_0..k) | H nats | |—|—|—|—:| | outer | add | [0.0, 0.0, 0.5, 0.5, 0.0, 0.0, 0.0, 0.0] | 0.693 | | outer | answer | [0.0, 0.0, 0.5, 0.5, 0.0, 0.0, 0.0, 0.0] | 0.693 | | outer | mul | [0.0, 0.0, 0.5, 0.5, 0.0, 0.0, 0.0, 0.0] | 0.693 | | outer | none | [0.0, 0.0, 0.5, 0.5, 0.0, 0.0, 0.0, 0.0] | 0.693 | | outer | sub | [0.0, 0.0, 0.5, 0.5, 0.0, 0.0, 0.0, 0.0] | 0.693 | | cos(update at outer scale, update at inner scale) | - | | cos(update_l, update_{l+1}) across OUTER depth | 0.18 | | cos(inner-scale update field at level l, same field at l+1) [the non-tautological cross-scale coherence] | - | | cos across INNER sub-depth inside one sub-cell | - | | frac of tokens whose sub-router kept the same top-2 across sub-levels | - |

Growth map on the trained checkpoint (RMS(x); outer axis and inner axis logged separately — nesting adds a scale axis along which divergence can occur): | outer level | RMS(x) after level | max inner RMS inside that level | |—:|—:|—:| | 0 | 1.03 | - | | 1 | 1.46 | - | | 2 | 2.03 | - | | 3 | 2.05 | - | | 4 | 2.21 | - | | 5 | 2.53 | - | | 6 | 3.29 | - | | 7 | 8.90 | - |

FR_flat_g16 (variant flat, n=400 test items, dataset calc_mix)

Settling depth (post-hoc headroom for early exit; eps_rel=0.1):

bucket	mean levels used (stream)	mean (router-only)	frac <=4 (stream)	n tokens
trivial	8.00	8.00	0.0%	16,435
easy	8.00	8.00	0.0%	20,045
medium	8.00	8.00	0.0%	5,527
hard	8.00	8.00	0.0%	3,385

Router entropy per scale: outer H = 3.154 nats (max 4.159); inner H = - nats

quantity	value
JSD(outer selection, trivial vs hard)	0.00094

quantity	value
JSD(inner selection, trivial vs hard)	-
JSD(outer selection, 1-digit vs max-digit operands) [carry proxy]	0.00238
JSD(inner selection, 1-digit vs max-digit operands) [carry proxy]	-

Selection distribution by nearest operator (does routing specialise by op type? uniform => no): | scale | op | p(expert_0..k) | H nats | |—|—|—|—:| | outer | add | [0.016, 0.013, 0.016, 0.016, 0.016, 0.015, 0.017, 0.015, 0.016, 0.013, 0.019, 0.013, 0.015, 0.012, 0.015, 0.016, 0.015, 0.012, 0.015, 0.015, 0.016, 0.015, 0.016, 0.015, 0.015, 0.014, 0.015, 0.014, 0.017, 0.015, 0.015, 0.018, 0.021, 0.02, 0.019, 0.013, 0.016, 0.013, 0.018, 0.012, 0.013, 0.021, 0.016, 0.013, 0.018, 0.014, 0.012, 0.013, 0.016, 0.018, 0.016, 0.015, 0.014, 0.016, 0.015, 0.014, 0.026, 0.014, 0.022, 0.018, 0.015, 0.012, 0.013, 0.018] | 4.146 | | outer | answer | [0.015, 0.019, 0.021, 0.022, 0.017, 0.016, 0.014, 0.018, 0.021, 0.017, 0.011, 0.017, 0.013, 0.017, 0.013, 0.013, 0.02, 0.02, 0.018, 0.019, 0.016, 0.021, 0.013, 0.016, 0.011, 0.014, 0.017, 0.013, 0.019, 0.024, 0.019, 0.017, 0.013, 0.012, 0.015, 0.02, 0.018, 0.023, 0.019, 0.021, 0.017, 0.015, 0.011, 0.02, 0.02, 0.017, 0.013, 0.022, 0.015, 0.023, 0.011, 0.015, 0.02, 0.018, 0.021, 0.013, 0.01, 0.012, 0.005, 0.011, 0.003, 0.006, 0.001, 0.002] | 4.097 | | outer | mul | [0.015, 0.015, 0.013, 0.009, 0.018, 0.014, 0.017, 0.014, 0.015, 0.011, 0.015, 0.017, 0.016, 0.012, 0.015, 0.019, 0.017, 0.014, 0.013, 0.017, 0.016, 0.016, 0.018, 0.021, 0.014, 0.016, 0.014, 0.017, 0.013, 0.013, 0.017, 0.021, 0.018, 0.017, 0.014, 0.017, 0.018, 0.013, 0.014, 0.013, 0.014, 0.014, 0.015, 0.011, 0.013, 0.013, 0.017, 0.012, 0.019, 0.017, 0.014, 0.016, 0.016, 0.018, 0.015, 0.011, 0.022, 0.013, 0.026, 0.019, 0.019, 0.013, 0.015, 0.021] | 4.141 | | outer | none | [0.023, 0.01, 0.008, 0.022, 0.012, 0.013, 0.012, 0.003, 0.009, 0.016, 0.019, 0.013, 0.023, 0.016, 0.017, 0.016, 0.015, 0.013, 0.02, 0.005, 0.021, 0.011, 0.012, 0.005, 0.033, 0.038, 0.02, 0.034, 0.005, 0.011, 0.006, 0.006, 0.019, 0.013, 0.015, 0.009, 0.009, 0.011, 0.02, 0.005, 0.012, 0.016, 0.012, 0.007, 0.007, 0.018, 0.022, 0.018, 0.01, 0.017, 0.037, 0.02, 0.018, 0.009, 0.005, 0.023, 0.012, 0.031, 0.01, 0.022, 0.021, 0.019, 0.018, 0.026] | 4.034 | | outer | sub | [0.016, 0.014, 0.016, 0.016, 0.018, 0.015, 0.017, 0.015, 0.016, 0.013, 0.02, 0.014, 0.014, 0.012, 0.015, 0.015, 0.015, 0.013, 0.015, 0.015, 0.016, 0.015, 0.016, 0.015, 0.015, 0.014, 0.016, 0.014, 0.016, 0.016, 0.015, 0.018, 0.02, 0.019, 0.019, 0.014, 0.015, 0.012, 0.02, 0.013, 0.013, 0.02, 0.016, 0.014, 0.017, 0.014, 0.011, 0.013, 0.018, 0.017, 0.016, 0.013, 0.013, 0.016, 0.015, 0.015, 0.026, 0.014, 0.022, 0.019, 0.015, 0.012, 0.013, 0.017] | 4.147 | | cos(update at outer scale, update at inner scale) | - | | cos(update_l, update_{l+1}) across OUTER depth | 0.0582 | | cos(inner-scale update field at level l, same field at l+1) [the non-tautological cross-scale coherence] | - | | cos across INNER sub-depth inside one sub-cell | - | | frac of tokens whose sub-router kept the same top-2 across sub-levels | - |

Growth map on the trained checkpoint (RMS(x); outer axis and inner axis logged separately — nesting adds a scale axis along which divergence can occur): | outer level | RMS(x) after level | max inner RMS inside that level | |—:|—:|—:| | 0 | 1.09 | - | | 1 | 1.66 | - | | 2 | 2.22 | - | | 3 | 2.44 | - | | 4 | 2.46 | - | | 5 | 2.39 | - | | 6 | 3.96 | - | | 7 | 10.42 | - |

FR_flat_g16_s1 (variant flat, n=400 test items, dataset calc_mix)

Settling depth (post-hoc headroom for early exit; eps_rel=0.1):

bucket	mean levels used (stream)	mean (router-only)	frac <=4 (stream)	n tokens
trivial	8.00	8.00	0.0%	16,435
easy	8.00	8.00	0.0%	20,045
medium	8.00	8.00	0.0%	5,527

bucket	mean levels used (stream)	mean (router-only)	frac <=4 (stream)	n tokens
hard	8.00	8.00	0.0%	3,385

Router entropy per scale: outer H = 2.939 nats (max 4.159); inner H = - nats

quantity	value
JSD(outer selection, trivial vs hard)	0.00077
JSD(inner selection, trivial vs hard)	-
JSD(outer selection, 1-digit vs max-digit operands) [carry proxy]	0.00132
JSD(inner selection, 1-digit vs max-digit operands) [carry proxy]	-

Selection distribution by nearest operator (does routing specialise by op type? uniform => no): | scale | op | p(expert_0..k) | H nats | |—|—|—|—:| | outer | add | [0.019, 0.013, 0.019, 0.018, 0.013, 0.011, 0.018, 0.013, 0.014, 0.016, 0.012, 0.022, 0.013, 0.018, 0.014, 0.015, 0.018, 0.017, 0.017, 0.016, 0.014, 0.016, 0.016, 0.015, 0.01, 0.018, 0.014, 0.017, 0.016, 0.015, 0.016, 0.019, 0.011, 0.017, 0.015, 0.018, 0.016, 0.013, 0.016, 0.015, 0.018, 0.016, 0.011, 0.019, 0.018, 0.015, 0.021, 0.012, 0.013, 0.017, 0.014, 0.015, 0.015, 0.014, 0.018, 0.012, 0.016, 0.02, 0.011, 0.014, 0.017, 0.017, 0.017, 0.019] | 4.145 | | outer | answer | [0.011, 0.02, 0.02, 0.015, 0.016, 0.018, 0.012, 0.017, 0.023, 0.017, 0.018, 0.017, 0.015, 0.01, 0.019, 0.015, 0.016, 0.022, 0.023, 0.016, 0.016, 0.021, 0.016, 0.016, 0.02, 0.021, 0.012, 0.017, 0.018, 0.016, 0.01, 0.011, 0.017, 0.02, 0.015, 0.023, 0.017, 0.015, 0.021, 0.018, 0.02, 0.017, 0.016, 0.016, 0.014, 0.016, 0.017, 0.014, 0.018, 0.019, 0.014, 0.013, 0.018, 0.025, 0.011, 0.015, 0.011, 0.016, 0.014, 0.005, 0.005, 0.004, 0.002, 0.002] | 4.102 | | outer | mul | [0.019, 0.01, 0.019, 0.019, 0.013, 0.008, 0.017, 0.01, 0.009, 0.014, 0.013, 0.017, 0.014, 0.018, 0.013, 0.013, 0.015, 0.016, 0.012, 0.018, 0.012, 0.015, 0.012, 0.02, 0.015, 0.013, 0.015, 0.013, 0.017, 0.017, 0.019, 0.019, 0.012, 0.016, 0.02, 0.015, 0.012, 0.011, 0.016, 0.016, 0.019, 0.013, 0.009, 0.019, 0.023, 0.018, 0.018, 0.011, 0.015, 0.013, 0.016, 0.019, 0.017, 0.012, 0.022, 0.012, 0.017, 0.018, 0.012, 0.017, 0.019, 0.017, 0.023, 0.024] | 4.132 | | outer | none | [0.017, 0.016, 0.017, 0.008, 0.015, 0.012, 0.013, 0.011, 0.005, 0.009, 0.011, 0.0, 0.01, 0.023, 0.007, 0.022, 0.018, 0.012, 0.022, 0.02, 0.008, 0.017, 0.023, 0.012, 0.019, 0.009, 0.019, 0.015, 0.013, 0.02, 0.02, 0.025, 0.008, 0.009, 0.008, 0.01, 0.007, 0.019, 0.022, 0.014, 0.003, 0.022, 0.024, 0.01, 0.007, 0.015, 0.003, 0.022, 0.021, 0.008, 0.022, 0.023, 0.033, 0.002, 0.014, 0.044, 0.017, 0.01, 0.019, 0.018, 0.03, 0.019, 0.021, 0.03] | 4.023 | | outer | sub | [0.02, 0.013, 0.02, 0.019, 0.013, 0.011, 0.019, 0.014, 0.013, 0.015, 0.01, 0.022, 0.012, 0.017, 0.014, 0.015, 0.02, 0.017, 0.016, 0.016, 0.016, 0.017, 0.015, 0.016, 0.01, 0.019, 0.014, 0.016, 0.016, 0.015, 0.017, 0.02, 0.011, 0.018, 0.014, 0.018, 0.017, 0.012, 0.015, 0.015, 0.019, 0.015, 0.011, 0.019, 0.018, 0.015, 0.02, 0.012, 0.013, 0.015, 0.014, 0.014, 0.015, 0.014, 0.018, 0.011, 0.016, 0.02, 0.011, 0.014, 0.016, 0.017, 0.017, 0.019] | 4.142 | | cos(update at outer scale, update at inner scale) | - | | cos(update_l, update_{l+1}) across OUTER depth | 0.0777 | | cos(inner-scale update field at level l, same field at l+1) [the non-tautological cross-scale coherence] | - | | cos across INNER sub-depth inside one sub-cell | - | | frac of tokens whose sub-router kept the same top-2 across sub-levels | - |

Growth map on the trained checkpoint (RMS(x); outer axis and inner axis logged separately — nesting adds a scale axis along which divergence can occur): | outer level | RMS(x) after level | max inner RMS inside that level | |—:|—:|—:| | 0 | 1.13 | - | | 1 | 1.71 | - | | 2 | 2.04 | - | | 3 | 2.43 | - | | 4 | 2.61 | - | | 5 | 2.82 | - | | 6 | 3.35 | - | | 7 | 7.46 | - |

FR_flat_g4 (variant flat, n=400 test items, dataset calc_mix)

Settling depth (post-hoc headroom for early exit; eps_rel=0.1):

bucket	mean levels used (stream)	mean (router-only)	frac <=4 (stream)	n tokens
trivial	8.00	8.00	0.0%	16,435
easy	8.00	8.00	0.0%	20,045
medium	8.00	8.00	0.0%	5,527
hard	8.00	8.00	0.0%	3,385

Router entropy per scale: outer H = 1.908 nats (max 2.773); inner H = - nats

quantity	value
JSD(outer selection, trivial vs hard)	0.00096
JSD(inner selection, trivial vs hard)	-
JSD(outer selection, 1-digit vs max-digit operands) [carry proxy]	0.00198
JSD(inner selection, 1-digit vs max-digit operands) [carry proxy]	-

Selection distribution by nearest operator (does routing specialise by op type? uniform => no): | scale | op | p(expert_0..k) | H nats | |—|—|—|—:| | outer | add | [0.064, 0.067, 0.064, 0.085, 0.072, 0.057, 0.056, 0.046, 0.055, 0.054, 0.039, 0.068, 0.046, 0.087, 0.071, 0.07] | 2.751 | | outer | answer | [0.065, 0.073, 0.08, 0.075, 0.064, 0.065, 0.077, 0.091, 0.089, 0.068, 0.061, 0.067, 0.07, 0.023, 0.021, 0.011] | 2.688 | | outer | mul | [0.062, 0.064, 0.042, 0.089, 0.056, 0.058, 0.063, 0.047, 0.06, 0.061, 0.042, 0.064, 0.062, 0.089, 0.061, 0.081] | 2.750 | | outer | none | [0.074, 0.05, 0.054, 0.061, 0.065, 0.035, 0.07, 0.03, 0.058, 0.036, 0.105, 0.081, 0.044, 0.058, 0.09, 0.088] | 2.718 | | outer | sub | [0.069, 0.068, 0.065, 0.086, 0.068, 0.058, 0.059, 0.045, 0.061, 0.052, 0.035, 0.068, 0.041, 0.087, 0.073, 0.066] | 2.747 | | cos(update at outer scale, update at inner scale) | - | | cos(update_l, update_{l+1}) across OUTER depth | 0.115 | | cos(inner-scale update field at level l, same field at l+1) [the non-tautological cross-scale coherence] | - | | cos across INNER sub-depth inside one sub-cell | - | | frac of tokens whose sub-router kept the same top-2 across sub-levels | - |

Growth map on the trained checkpoint (RMS(x); outer axis and inner axis logged separately — nesting adds a scale axis along which divergence can occur): | outer level | RMS(x) after level | max inner RMS inside that level | |—:|—:|—:| | 0 | 1.45 | - | | 1 | 1.92 | - | | 2 | 2.28 | - | | 3 | 2.72 | - | | 4 | 3.40 | - | | 5 | 4.24 | - | | 6 | 4.80 | - | | 7 | 8.88 | - |

FR_flat_long (variant flat, n=400 test items, dataset calc_mix)

Settling depth (post-hoc headroom for early exit; eps_rel=0.1):

bucket	mean levels used (stream)	mean (router-only)	frac <=4 (stream)	n tokens
trivial	7.92	7.87	0.0%	16,435
easy	7.93	7.88	0.0%	20,045
medium	7.98	7.96	0.0%	5,527
hard	8.00	7.99	0.0%	3,385

Router entropy per scale: outer H = 1.281 nats (max 2.079); inner H = - nats

quantity	value
JSD(outer selection, trivial vs hard)	0.00091
JSD(inner selection, trivial vs hard)	-
JSD(outer selection, 1-digit vs max-digit operands) [carry proxy]	0.00140
JSD(inner selection, 1-digit vs max-digit operands) [carry proxy]	-

Selection distribution by nearest operator (does routing specialise by op type? uniform => no): | scale | op | p(expert_0..k) | H nats | |—|—|—|—:| | outer | add | [0.199, 0.168, 0.146, 0.085, 0.082, 0.107, 0.093, 0.12] | 2.031 | | outer | answer | [0.143, 0.142, 0.111, 0.15, 0.139, 0.13, 0.12, 0.066] | 2.056 | | outer | mul | [0.179, 0.156, 0.154, 0.091, 0.088, 0.104, 0.09, 0.138] | 2.043 | | outer | none | [0.147, 0.167, 0.111, 0.091, 0.062, 0.103, 0.137, 0.181] | 2.032 | | outer | sub | [0.193, 0.17, 0.128, 0.097, 0.095, 0.112, 0.094, 0.111] | 2.043 | | cos(update at outer scale, update at inner scale) | - | | cos(update_l, update_{l+1}) across OUTER depth | 0.1433 | | cos(inner-scale update field at level l, same field at l+1) [the non-tautological cross-scale coherence] | - | | cos across INNER sub-depth inside one sub-cell | - | | frac of tokens whose sub-router kept the same top-2 across sub-levels | - |

Growth map on the trained checkpoint (RMS(x); outer axis and inner axis logged separately — nesting adds a scale axis along which divergence can occur): | outer level | RMS(x) after level | max inner RMS inside that level | |—:|—:|—:| | 0 | 1.60 | - | | 1 | 2.32 | - | | 2 | 2.73 | - | | 3 | 3.54 | - | | 4 | 5.01 | - | | 5 | 5.31 | - | | 6 | 6.55 | - | | 7 | 10.57 | - |

FR_flat_s1 (variant flat, n=400 test items, dataset calc_mix)

Settling depth (post-hoc headroom for early exit; eps_rel=0.1):

bucket	mean levels used (stream)	mean (router-only)	frac <=4 (stream)	n tokens
trivial	7.99	7.99	0.0%	16,435
easy	7.99	7.98	0.0%	20,045
medium	8.00	7.97	0.0%	5,527
hard	8.00	7.97	0.0%	3,385

Router entropy per scale: outer H = 0.999 nats (max 2.079); inner H = - nats

quantity	value
JSD(outer selection, trivial vs hard)	0.00067
JSD(inner selection, trivial vs hard)	-
JSD(outer selection, 1-digit vs max-digit operands) [carry proxy]	0.00170
JSD(inner selection, 1-digit vs max-digit operands) [carry proxy]	-

Selection distribution by nearest operator (does routing specialise by op type? uniform => no): | scale | op | p(expert_0..k) | H nats | |—|—|—|—:| | outer | add | [0.183, 0.185, 0.022, 0.106, 0.111, 0.135,

0.112, 0.147] | 1.984 | | outer | answer | [0.126, 0.143, 0.171, 0.127, 0.162, 0.114, 0.094, 0.063] | 2.042 | | outer | mul | [0.166, 0.181, 0.018, 0.102, 0.112, 0.148, 0.119, 0.154] | 1.982 | | outer | none | [0.2, 0.2, 0.038, 0.115, 0.181, 0.103, 0.145, 0.018] | 1.913 | | outer | sub | [0.181, 0.185, 0.02, 0.109, 0.109, 0.139, 0.114, 0.142] | 1.982 | | cos(update at outer scale, update at inner scale) | - | | cos(update_l, update_{l+1}) across OUTER depth | 0.1458 | | cos(inner-scale update field at level l, same field at l+1) [the non-tautological cross-scale coherence] | - | | cos across INNER sub-depth inside one sub-cell | - | | frac of tokens whose sub-router kept the same top-2 across sub-levels | - |

Growth map on the trained checkpoint (RMS(x); outer axis and inner axis logged separately — nesting adds a scale axis along which divergence can occur): | outer level | RMS(x) after level | max inner RMS inside that level | |—:|—:|—:| | 0 | 1.55 | - | | 1 | 1.93 | - | | 2 | 2.60 | - | | 3 | 3.01 | - | | 4 | 3.26 | - | | 5 | 4.22 | - | | 6 | 5.73 | - | | 7 | 9.19 | - | ### T4a. Teacher-forced competence vs autoregressive exactness

Caption: test split of calc_mix, n as in each row, greedy decoding. `tf_acc_trace` is teacher-forced top-1 on trace positions (the hard part); prefix survival is the fraction of items whose rollout still matches the target after k emitted tokens; `frac_no_divergence` = rollouts that matched the entire target.

run	tf_acc_prompt	tf_acc_trace	tf loss	surv@8	surv@16	surv@32	never diverged	exact (full trace)	n
FR_A	27.58%	95.94%	0.459	99.00%	83.00%	70.00%	~43% (bucket-mean)	65.50%	200
FR_A_g16	27.23%	95.84%	0.459	97.75%	79.25%	67.75%	~42% (bucket-mean)	64.50%	400
FR_A_long	28.48%	98.27%	0.410	99.50%	89.00%	82.50%	~61% (bucket-mean)	79.00%	200
FR_B	27.64%	92.41%	0.527	98.00%	68.00%	55.50%	~36% (bucket-mean)	55.50%	200
FR_flat	28.96%	94.00%	0.503	97.50%	78.00%	64.50%	~41% (bucket-mean)	64.00%	200
FR_flat1b	28.18%	92.50%	0.530	97.50%	67.00%	55.50%	~36% (bucket-mean)	55.50%	200
FR_flat_g16	28.00%	95.29%	0.475	98.00%	75.00%	59.50%	~36% (bucket-mean)	56.50%	200
FR_flat_g4	28.30%	94.40%	0.492	100.00%	76.00%	65.50%	~41% (bucket-mean)	65.00%	200
FR_flat_long	28.72%	96.62%	0.449	99.50%	87.50%	76.50%	~49% (bucket-mean)	73.00%	200

T6. Matched-step trajectories (secondary; same steps, not different budgets)

step	FR_flat	FR_flat1b	FR_A	FR_B	FR_flat_g4	FR_flat_g16	FR_A_g16
1,500	36.50% (loss 0.797)	32.80% (loss 0.868)	34.50% (loss 0.815)	32.30% (loss 0.940)	37.30% (loss 0.823)	28.30% (loss 0.830)	36.80% (loss 0.850)
3,000	46.90% (loss 0.647)	43.10% (loss 0.697)	44.80% (loss 0.634)	40.10% (loss 0.686)	45.60% (loss 0.676)	39.10% (loss 0.668)	46.10% (loss 0.667)
4,500	52.50% (loss 0.603)	47.50% (loss 0.628)	49.20% (loss 0.587)	46.30% (loss 0.634)	49.10% (loss 0.606)	47.60% (loss 0.588)	51.50% (loss 0.586)
6,000	55.20% (loss 0.564)	49.40% (loss 0.593)	52.80% (loss 0.544)	48.70% (loss 0.600)	54.50% (loss 0.558)	52.50% (loss 0.532)	55.30% (loss 0.532)
7,500	56.90% (loss 0.545)	51.10% (loss 0.566)	56.60% (loss 0.515)	50.50% (loss 0.576)	56.50% (loss 0.540)	53.30% (loss 0.500)	57.00% (loss 0.500)
9,000	58.50% (loss 0.541)	51.90% (loss 0.554)	58.60% (loss 0.496)	52.50% (loss 0.561)	57.00% (loss 0.530)	53.80% (loss 0.486)	59.70% (loss 0.484)

T4. Forced-depth curves and oracle bound

Oracle set: G=8 pair (9,000 steps)

- **CONDITION 1 flat top-2 MoE** (run `FR_flat`, budget 9000 steps, n=400 test items, dataset calc_mix): oracle ceiling **61.10%** at 60,349,623 MACs/item; best fixed-depth 61.10%; monotone in depth = True; depths={'trivial': 8, 'easy': 8, 'medium': 8, 'hard': 8}
- **CONDITION 3 hierarchical fractal** (run `FR_A`, budget 9000 steps, n=400 test items, dataset calc_mix): oracle ceiling **61.25%** at 60,009,337 MACs/item; best fixed-depth 61.25%; monotone in depth = True; depths={'trivial': 8, 'easy': 8, 'medium': 8, 'hard': 8}
- **CONDITION 3B strict cross-scale sharing** (run `FR_B`, budget 9000 steps, n=400 test items, dataset calc_mix): oracle ceiling **54.82%** at 60,009,337 MACs/item; best fixed-depth 54.82%; monotone in depth = True; depths={'trivial': 8, 'easy': 8, 'medium': 8, 'hard': 8}

  -> Condition 3 oracle ceiling 61.25% sits ABOVE the flat model's best fixed-depth accuracy 61.10% at matched budget.

Oracle set: G=16 pair (9,000 steps)

- **CONDITION 1 flat top-2 MoE** (run `FR_flat_g16`, budget 9000 steps, n=400 test items, dataset calc_mix): oracle ceiling **56.08%** at 58,806,995 MACs/item; best fixed-depth 56.08%; monotone in depth = True; depths={'trivial': 8, 'easy': 8, 'medium': 8, 'hard': 8}
- **CONDITION 3 hierarchical fractal** (run `FR_A_g16`, budget 9000 steps, n=400 test items, dataset calc_mix): oracle ceiling **63.66%** at 59,714,423 MACs/item; best fixed-depth 63.66%; monotone in depth = True; depths={'trivial': 8, 'easy': 8, 'medium': 8, 'hard': 8}

  -> Condition 3 oracle ceiling 63.66% sits ABOVE the flat model's best fixed-depth accuracy 56.08% at matched budget.

Oracle set: long-budget pair (18,000 steps)

- **CONDITION 1 flat top-2 MoE** (run `FR_flat_long`, budget 18000 steps, n=400 test items, dataset calc_mix): oracle ceiling **68.49%** at 60,349,623 MACs/item; best fixed-depth 68.49%; monotone in depth = True; depths={'trivial': 8, 'easy': 8, 'medium': 8, 'hard': 8}
- **CONDITION 3 hierarchical fractal** (run `FR_A_long`, budget 18000 steps, n=400 test items, dataset calc_mix): oracle ceiling **72.27%** at 60,009,337 MACs/item; best fixed-depth 72.27%; monotone in depth = True; depths={'trivial': 8, 'easy': 8, 'medium': 8, 'hard': 8}

  -> Condition 3 oracle ceiling 72.27% sits ABOVE the flat model's best fixed-depth accuracy 68.49% at matched budget.

CONDITION 1 flat top-2 MoE (run FR_flat_long, budget 18000 steps, n=400 test items)

outer depth	aggregate exact
1	0.00%
2	0.00%
3	0.00%
4	0.48%
6	34.01%
8	68.49%

Oracle ceiling (per-bucket best depth): **68.49%** at 60,349,623 MACs/item, depths={'trivial': 8, 'easy': 8, 'medium': 8, 'hard': 8}, monotone in depth = True.

target	reachable	min MACs/item	achieved acc	depths
50%	yes	43,311,508	55.76%	{'trivial': 6, 'easy': 8, 'medium': 1, 'hard': 1}
55%	yes	43,311,508	55.76%	{'trivial': 6, 'easy': 8, 'medium': 1, 'hard': 1}
60%	yes	48,585,569	65.88%	{'trivial': 8, 'easy': 8, 'medium': 1, 'hard': 1}
65%	yes	48,585,569	65.88%	{'trivial': 8, 'easy': 8, 'medium': 1, 'hard': 1}
70%	no	-	-	-

CONDITION 3 hierarchical fractal (run FR_A_long, budget 18000 steps, n=400 test items)

outer depth	aggregate exact
1	0.00%
2	0.00%
3	0.72%
4	1.26%
6	16.16%
8	72.27%

Oracle ceiling (per-bucket best depth): **72.27%** at 60,009,337 MACs/item, depths={'trivial': 8, 'easy': 8, 'medium': 8, 'hard': 8}, monotone in depth = True.

target	reachable	min MACs/item	achieved acc	depths
50%	yes	48,311,623	65.88%	{'trivial': 8, 'easy': 8, 'medium': 1, 'hard': 1}
55%	yes	48,311,623	65.88%	{'trivial': 8, 'easy': 8, 'medium': 1, 'hard': 1}
60%	yes	48,311,623	65.88%	{'trivial': 8, 'easy': 8, 'medium': 1, 'hard': 1}
65%	yes	48,311,623	65.88%	{'trivial': 8, 'easy': 8, 'medium': 1, 'hard': 1}
70%	yes	55,777,579	72.27%	{'trivial': 8, 'easy': 8, 'medium': 8, 'hard': 1}

7. Verdict

Headline: scale recursion helps only where granularity is forced, and content-dependent routing — the mechanism the hypothesis named — subtracts accuracy in BOTH families. At matched parameters, matched active parameters per token and matched MACs/item (all three within 0.7%, T1), nested and flat tie at coarse granularity — **58.60% exact / 59.80% answer** (Condition 3A) vs **58.50% / 59.30%** (Condition 1); calc_mix val split, n=1,000 items, 9,000 steps, seed 0, best-by-exact; McNemar on the same 500 test items p = 0.72 (17 vs 14 discordant). At seed 1 that same pair *reverses* and favours flat (−4.6 points, p = 2.3e-05), so no superiority over a well-configured flat MoE survives replication. What does survive is narrower: nesting beats any flat model forced to the *same fine granularity*, at both rungs and both seeds.

Every row names both denominators: **val** = best-by-exact on 1,000 calc_mix val items (the T1 checkpoint rule); **test** = greedy answer accuracy on the same 500 calc_mix test items, which is what McNemar runs on. The two are not interchangeable — the test split has a different bucket mix — so both are printed rather than merged into one cell.

comparison (params / active-per-token / MACs-item matched)	flat val	nested val	flat test	nested test	discordant flat-only/ nested-only	p
coarse: flat G=2 vs nested G=8, 9k steps	58.50%	58.60%	61.60%	61.00%	17 / 14	0.72
mid: flat G=4 vs nested G=8, 9k steps	57.00%	58.60%	61.60%	61.00%	18 / 15	0.73
same granularity G=8: flat E=32/e=31/ top-8 vs nested	**51.90%**	**58.60%**	52.80%	61.00%	4 / 45	**1.1e-08**
fine G=16: flat E=64/ e=15/top-16 vs nested G=16	**53.80%**	**59.70%**	55.20%	63.20%	8 / 48	**1.9e-07**
longer budget: flat G=2 vs nested G=8, **18k steps**	**64.80%**	**71.20%**	67.40%	72.20%	13 / 37	**0.0011**

(1) Nesting buys fine granularity; it does not beat coarse routing. Splitting the same active width into many narrow experts costs a flat model 4.7–6.6 points — val 58.50% at G=2 → 57.00% at G=4 → 51.90% at G=8, partially recovering to 53.80% at G=16 (T8; a clear drop, though not strictly monotone) — while the nested model at exactly the same leaf count, leaf width, active width units per level (248 at G=8, 240 at G=16) and MACs/item *gains*: 58.60% at G=8 and **59.70%** at G=16. `FR_flat1b` vs `FR_A` is the cleanest contrast in the study: identical 8 executed leaves of width 31 per outer level, identical 64 sequential FFN compositions per token (T7), differing only in that the nested model's second level of selection lives **inside** the expert its first level chose (5 routers/token vs 1). That single difference is worth **+6.7 val points at seed 0 and +3.7 at seed 1**; paired on the 500-item test split, 61.00% vs 52.80% ($p = 1.1e\text{-}08$) at seed 0 and 57.60% vs 54.60% ($p = 0.015$) at seed 1. So the answer to "does nesting beat flat at higher granularity?" is **yes, at matched granularity and at both seeds**: +6.7/+3.7 val at G=8 and +5.9/+8.8 at G=16 (paired $p \le 0.015$ in all four cells). The answer to "does nesting beat the *best* flat model?" is **no**: against coarse flat G=2 the nested model is −0.6 test points ($p = 0.72$) at seed 0 and −4.6 ($p = 2.3e\text{-}05$) at seed 1. The nested model's own accuracy rises with granularity while the flat model's falls, but a flat model simply *allowed to stay coarse* is never beaten.

(2) Longer training does NOT change the verdict — the seed-0 long run did not replicate. At 18,000 steps and seed 0 the nested model reached **71.20% exact** / **72.00% answer** vs flat **64.80%** / **66.20%** (p = 0.0011). Rerun at seed 1 with identical settings the sign flips: flat **63.20% val** / **68.20% test** vs nested **62.30%** / **66.40%**, McNemar p = 0.18 (22 vs 13 discordant, flat ahead). The honest reading is that the long-budget pair is seed-dominated and swings by ~7 points in either direction; "the advantage grows with budget" is **not supported**. This also means the previous report's Pareto-dominance claim should not be re-used as a general statement about routed models: it was measured on a looped architecture at one budget, and here the same-budget tie does not survive doubling that budget.

(2b) Seed replication changes part of the story, and it must be reported. All headline comparisons were rerun at seed 1 (T11 + `research/results/paired_*_s1*.json`, n = 500 shared items, McNemar with continuity correction):

comparison (paired, n=500)	s0 flat	s0 nested	s0 p	s1 flat	s1 nested	s1 p
same granularity G=8	52.8	61.0	1.1e-08	54.6	57.6	**0.015**
fine granularity G=16	55.2	63.2	1.9e-07	48.0	57.6	**2.4e-09**
coarse flat G=2 vs nested G=8	61.6	61.0	0.72	**62.2**	**57.6**	**2.3e-05 (flat wins)**
nested routed vs nested fixed-inner	66.0	61.0	1.2e-04	60.4	57.6	**0.018**

Two claims replicate and one does not. **Replicates:** nesting beats a flat model *at the same granularity* at both rungs and both seeds (+3.0 to +9.6 points), and removing the inner routers improves the nested model at both seeds (+2.8, +5.0). **Does not replicate:** "nesting beats the best coarse-granularity flat model". That pair was a tie at seed 0 (−0.6, p = 0.72) and flips to a *significant flat win* at seed 1 (−4.6, p = 2.3e-05). The correct claim is therefore narrower than the one this section opened with: **at 9,000 steps nesting does not reliably beat the best flat model available; it reliably beats any flat model forced to the same fine granularity — nesting buys fine granularity rather than beating coarse routing.** The 18k-step seed-0 pair (+6.4 points, p = 0.0011) also failed to replicate: at seed 1 it is −1.8 points in flat's favour (p = 0.18), see (2).

(2c) The reciprocal control kills the hierarchy explanation of `FR_A_fixed`. Dropping the inner routers and averaging a *fixed* random subset of sub-experts lifted the nested model by +3.0 val points (61.60% vs 58.60%) / +5.0 test points (66.00% vs 61.00%, p = 1.2e-04). That is only evidence about hierarchy if the same trick leaves the flat family alone — it does not, and it helps there *more* (T12; test-split accuracies from the paired files, val deltas in the last column):

fixed-subset averaging vs its routed twin	seed 0 test	routed test	delta (test)	p	delta (val)
	64.8%	61.6%	+3.2	0.008	+2.2

fixed-subset averaging vs its routed twin	seed 0 test	routed test	delta (test)	p	delta (val)
flat G=2 (E=16/ e=64/top-2) → `FR_flat_fixed`					
flat G=8 (E=32/ e=31/top-8) → `FR_flat1b_fixed`	**62.6%**	52.8%	**+9.8**	1.8e-11	+8.0
nested G=8 → `FR_A_fixed`	66.0%	61.0%	+5.0	1.2e-04	+3.0
nested G=16 → `FR_A_g16_fixed`	61.2%	63.2%	−2.0	0.089	−0.1

Content-independent subset averaging is worth up to +9.8 points to a *flat* MoE — more than the +5.0 it buys nested. At matched granularity the fixed flat model (G=8, 62.6%) still loses to the fixed nested model (66.0%, p = 0.01), so nesting retains its +3.4-point matched-granularity advantage after routing is removed from both; but the *global best* configuration in this whole study is a flat model with a fixed random subset, and the nested model with routers (`FR_A`, 61.0%) loses to it by 3.8 points (p = 0.24 for `FR_flat1b_fixed` vs `FR_A`). The seed-0 advantage of fixed-nested over fixed-flat (66.0% vs 62.6%, p = 0.01) **also fails to replicate**: at seed 1 it inverts, fixed-flat 65.8% vs fixed-nested 60.4% (p = 3.1e-05). So across the two seeds and both granularities there is no stable hierarchy effect once routing is removed from both families. **Routing is the thing that hurts, in both families; nesting's remaining value after routing is removed is seed noise.**

(3) The routing measurements do NOT support the mechanism the hypothesis named. The fractal story predicts hierarchical specialisation — outer router broad, sub-router specific — and early settling on easy items. Neither is present:

quantity (test split, n=400 items)	flat G=2	nested G=8	flat G=16	nested G=16
outer router entropy, normalised by log E	0.661	0.258	0.758	0.264
inner router entropy, normalised by log E_inner	—	0.431	—	0.286
JSD of outer selection, trivial vs hard bucket	0.00113	0.00075	0.00094	0.00055
JSD of inner selection, trivial vs hard	—	0.000095	—	0.000039
JSD, 1-digit vs max-digit	0.00462 / —	0.00019 / 0.00115	0.00238 / —	0.00052 / 0.00035

quantity (test split, n=400 items)	flat G=2	nested G=8	flat G=16	nested G=16
operands (carry proxy), outer / inner				
settling depth, router-only definition (trivial / easy / medium / hard)	7.9 / 7.85 / 7.89 / 7.98	7.38 / 7.13 / 7.36 / 7.79	8.0 / 8.0 / 8.0 / 8.0	7.59 / 7.67 / 7.67 / 7.80

The nested routers *are* sharper (normalised entropy 0.26 outer, 0.43 inner, versus 0.66–0.76 for flat pools), but sharpness is not content-selectivity: difficulty-conditioned selection divergence is $\leq$ 0.0011 nats in every model at every scale, and the sub-router's JSD across difficulty buckets (0.000095) is *smaller* than its parent's — i.e. if anything the inner scale is **less** difficulty-aware, not more. Operator-type and digit-position conditioning are equally flat (T3). Settling depth is 7.1–8.0 of 8 outer levels everywhere: there is no early settling on easy items, in either family, so the "easy tokens stop early" mechanism simply does not exist here. The nested model's advantage therefore did **not** come from difficulty-aware hierarchical routing. Combined with (2c), where fixed-subset averaging beats routed routing in *both* families, the conclusion is sharper than "the architecture wins and the explanation fails": **at this scale neither content- dependent routing nor nesting buys anything that a coarse flat MoE with uniform subset averaging does not already have.** The routers are the negative result; they cost 3–10 points wherever they are put.

(4) Coherence across scales is low, exactly as the fractal claim requires — and this is the one prediction that survived. `cos(update_l, update_{l+1})` across outer depth is **0.093** for the nested model (0.133 flat G=2, 0.058 flat G=16); the non-tautological cross-scale version — cosine between the *inner-scale* update field at outer level `l` and the same field at outer level `l+1`, on the tokens both scales touched — is **0.093** (nested G=8), 0.095 (G=16), 0.099 (18k steps); and cosine across *inner* sub-depth inside one sub-cell is **0.082**. All are far from the ≈ 0.85 that the weight-tied loop showed (FINDINGS §5.9). The scales do different work, which is what scale-recursion claims and what temporal tying could not achieve. One measurement in the data file must *not* be quoted: `cos(slot's returned delta, mean sub-level delta inside that slot)` = 0.9998 is ≈ 1 **by construction** (the returned delta *is* the sum of its sub-level deltas); it is recorded with a note to that effect so nobody mistakes it for evidence. Related: only 13–20% of tokens keep the same top-2 sub-experts across the two sub-levels of a sub-cell, so the inner routing genuinely re-decides instead of repeating — the opposite of the looped model's behaviour.

(5) Variant B (strict sharing across scale) reproduces the old failure, at a new axis. One sub-cell body shared by all 32 slots across all 8 outer levels, differentiated only by per-slot conditioning: **700,976 params (−81.3%)**, identical active params/token and MACs/item, **52.50% / 52.90%** — 6.1 points below Variant A. Sharing across scale buys parameters and costs accuracy while touching inference cost not at all: the same structural accounting that killed the depth-sharing variant, now confirmed on the scale axis. The §3.1 control explains why the per-slot conditioning is load-bearing rather than cosmetic: zero it and every slot computes an identical function (pairwise cos = 1.0000; swapping the router's selected slot changes the residual stream by 0.0000), so a naive strict fractal has an inert outer router *by construction*, before any training.

(6) Stability. Both nested variants pass the init-time gate with room to spare (worst RMS(x) 2.62 across outer **and inner** axes versus the 10^2 refusal threshold; T2), and neither produced a single non-finite step

in 9,000 or 18,000 steps (`ln_expert` is enforced by the constructor, so the §5 defect cannot be reintroduced). Trained growth stays bounded (RMS(x) 7.5–9.6 at the last outer level versus 292× in the pre-fix `D128_long`). Nesting did not add a divergence axis in practice — but note what it did add: the nested model is *less* depth-tolerant, reaching only 20.5% aggregate answer accuracy at half its outer depth where the flat G=2 model reaches 44.15% (T4). Capability is more concentrated in the last outer levels when experts are themselves routed cells, which shrinks exactly the adaptive-depth headroom the architecture was supposed to spend. The oracle question the task poses — does Condition 3's oracle ceiling sit above or below Condition 1's matched-budget accuracy? — answers itself per rung (T4, all curves monotone in depth, so each ceiling is its full-depth point): **tied** at G=8 / 9k steps (61.25% vs 61.10%), **above** at G=16 / 9k steps (**63.66%** vs 56.08%) and **above** at the long budget (**72.27%** vs 68.49%) — all of these seed-0 ceilings, so they inherit the seed-1 reversal reported in (2). There is no per-bucket depth headroom anywhere (every bucket's best depth is full depth, exactly as in the previous report), so nesting adds *no* adaptive-depth value; it raises the ceiling of the fixed-depth model instead. Variant B's ceiling is 54.82% at identical MACs/item.

Adaptive-depth verdict. Depth allocation remains dead for both families: monotone accuracy in outer depth, oracle depths (trivial, easy, medium, hard) = (8, 8, 8, 8) in every comparison, and the min-cost targets are reached only by dumping the medium/hard buckets to depth 1 — the accounting artefact flagged in FINDINGS §3. Nesting did not create the headroom it was partly designed to spend.

(6) Where the nested model's extra accuracy shows up in generation. Teacher-forced top-1 on trace positions is **95.94%** for `FR_A` vs 94.00% for `FR_flat` (tf loss 0.459 vs 0.503), and the advantage grows along the rollout rather than staying constant: prefix survival at 16 emitted tokens is **83.0%** vs 78.0%, at 32 tokens **70.0%** vs 64.5%, and the fraction of items whose rollout never diverges is ~43% vs ~41% (T4a, test split, n=200). At the long budget the same pattern is larger: 98.27% vs 96.62% teacher-forced, and 61% vs 49% of rollouts never diverging. So nesting reduces *error compounding during autoregressive decoding* on the arithmetic trace — a concrete generation-level signature, distinct from anything the routing hypothesis predicted. (`tf_acc_prompt` is ~28% for every model, nested and flat alike; it carries no discrimination signal here and is reported only so the trace-span denominator is explicit.)

(7) The control that settles the mechanism: removing the inner router makes the model *better* — and so does removing the outer one. `FR_A_fixed` keeps the nested topology, the leaf count, the leaf width and the composition depth, and replaces every sub-router with a **fixed random `k_inner`-subset average**, identical for all tokens — bit-identical parameter count (3,725,088) and MACs/item (60,009,337) to `FR_A`, verified by rebuilding both models. Result: **61.60% exact / 63.00% answer** vs `FR_A`'s 58.60% / 59.80% (val, n=1,000); paired on 500 test items **66.00% vs 61.00%**, discordant 32/7, **McNemar p = 1.2e-04** (seed 1: 60.4% vs 57.6%, p = 0.018). The sub-router is therefore not merely uninformative — it is *actively harmful*. Two caveats stated honestly: (a) the control changes two things at once, content-independence of the inner choice **and** the gate values (a plain ½+½ average instead of a renormalised softmax over the selected pair), so it should be read as "inner routing removed", not as a pure content-independence ablation; (b) averaging a fixed pair is an ensemble, and part of the gain may be ensembling rather than the absence of routing. Caveat (b) is now measured rather than assumed: the same trick applied to the **flat** family (T12, `FR_flat_fixed`, `FR_flat1b_fixed`) lifts it by +3.2 and **+9.8** val points, i.e. *more* than nesting ever earns, and makes fixed-flat the best configuration in the study at seed 0 (64.8%) and at seed 1 (65.8% vs fixed-nested 60.4%, p = 3.1e-05). **Fractal routing does not carry the benefit; neither does hierarchy — routing is the cost.**

Verdict. The fractal hypothesis — that scale-recursive structure allocates compute more efficiently than flat routing — is **not supported**. It survives only in its weakest form: nesting beats a flat MoE *forced to the same fine granularity* (+6.7/+3.0 val at G=8, +5.9/+9.6 at G=16, both seeds, $p \leq 0.015$). It fails everywhere it would matter: against a coarse flat model the nested model is −0.6 (p = 0.72) at seed 0 and

−4.6 (p = 2.3e-05) at seed 1; the 18k-step seed-0 win (+6.4, p = 0.0011) inverts to −1.8 (p = 0.18) at seed 1; and once routers are replaced by fixed random subset averaging in *both* families, the nesting advantage is +3.4 (p = 0.01) at seed 0 and **−5.4 (p = 3.1e-05)** at seed 1. Nothing hierarchical here replicates. The one robust finding of the study is about routing rather than about depth: **content-dependent softmax routing costs accuracy, in both families** — +3.2 (p = 0.008) and +9.8 (p = 1.8e-11) val points for removing it from flat models at G=2 and G=8, +5.0 (p = 1.2e-04) for removing it from the nested model — with no difficulty-, operator- or digit-position-conditioned specialisation anywhere to justify its cost (T3). The previous report's conclusion — weight tying is not compute sharing; sharing buys parameters, not MACs — remains correct and is *re-confirmed* here on the scale axis by Variant B. What changes relative to this paper's earlier draft is the claim that the topology wins: it does so at matched granularity only, and not across seeds. The honest name for what replicated is neither "fractal routing" nor "recursive width" but **granularity tolerance**: nesting makes a MoE insensitive to how finely its expert pool is cut, which is a trainability property, not an efficiency one.

Falsifying follow-ups. (i) *Is the gain composition depth or router count?* **Answered by this work: neither — it is recursive width, and routing subtracts.** The `FR_A_fixed` control (fixed random inner subsets, identical params/MACs/composition depth) beat the routed nested model by +3.0 val points, p = 1e-04. The follow-up that remains open is *which* part of that control did the work: content-independence of the inner choice, or the change in gate values (a plain ½+½ average rather than a renormalised softmax over the selected pair), or simple ensembling. **Partly answered since drafting:** the flat arm of this control was run (T12) and gained *more* than the nested arm, so "hierarchy" is out as an explanation; the arms that still separate the remaining candidates are fixed subsets with softmax gates, and learned-but-shared gates. If "fixed subsets + softmax gates" recovers `FR_A`'s worse result, the culprit is gate magnitude, not routing content. (ii) *Does the ladder keep favouring nesting?* Extend to G=32 both families at matched active width; nested must keep winning for the claim to be a scaling statement rather than one lucky rung. The seed-1 reversal of the coarse-granularity and fixed-vs-fixed pairs makes this more urgent than it looked: the matched-granularity wins replicated at both rungs, so G=32 is the rung that decides whether "granularity tolerance" is a scaling property or an artefact of two rungs. (iii) *Budget crossover*: **run and negative.** Both families at 18k steps, seed 0 favoured nesting by +6.4 (p = 0.0011); at seed 1 flat led by 1.8 (p = 0.18). The budget ladder does not survive replication, so "nesting pays more as budget grows" is withdrawn. (iv) *A task with genuine per-token compute heterogeneity*: calc_mix's difficulty is mostly trace length, a generation cost no router controls (FINDINGS §2.3). Multi-precision arithmetic or variable-width operand copying would give difficulty-conditioned routing something to bite on; if routing stays uniform there too, hierarchical routing has no mechanism on this task class and the "routing" half of the name should be dropped in favour of "recursive structure". (v) *Depth tolerance*: nesting made the model less tolerant of reduced outer depth (20.5% vs 44.15% at half depth). Training with inner depth randomized per token should flatten that curve; if it does not, concentration of capability in the final outer levels is intrinsic to nesting and caps any future adaptive-depth saving regardless of router quality.

8. Reproduction

```
# 1. matched-budget arithmetic (which axes can be matched, and to what tolerance)
PYTHONPATH=src python3 scripts/matched_hier.py

# 2. stability gate - refuses Condition 3 if init-time RMS(x) > 1e2 (outer AND inner axes)
PYTHONPATH=src python3 scripts/hier_init_blowup.py

# 3. the three trained conditions (9,000 steps, seed 0, batch 48, lr 5e-4, calc_mix)
C="--steps 9000 --batch 48 --lr 5e-4 --seed 0 --log-every 500 --eval-every 1500 --em-max 1000"
```

```
PYTHONPATH=src python3 scripts/train_hier.py --name FR_flat   --variant flat $C
PYTHONPATH=src python3 scripts/train_hier.py --name FR_flat1b --variant flat $C \
        --n-experts-flat 32 --e-hidden-flat 31 --k-outer 8      # granularity control
PYTHONPATH=src python3 scripts/train_hier.py --name FR_A      --variant A    $C
PYTHONPATH=src python3 scripts/train_hier.py --name FR_B      --variant B    $C

# 4. every measurement + the canonical tables
bash scripts/measure_all_hier.sh

# 5. wave 2: granularity ladder (flat G=4/G=16, nested G=16) + 18k-step reruns + paired tests
bash scripts/wave2.sh

# 6. wave 3: content-independent inner-routing control + seed-1 replicates
bash scripts/wave3.sh

# 7. wave 4/5: reciprocal fixed-subset controls on the FLAT family (--outer-route fixed),
#    fixed-vs-fixed across families, and the seed-1 replication of the 18k-step pair
bash scripts/wave4.sh && bash scripts/wave5.sh

# 8. final measurement pass over every run, then tables (pass EVERY run id, including _s1 and
#    _fixed variants, or T11/T12 emit empty rows)
R=$(ls -d runs/FR_* | sed 's|runs/||' | tr '\n' ',' | sed 's/,$//')
PYTHONPATH=src python3 scripts/hier_tables.py --runs "$R"

# 9. assemble the paper from its parts + generated tables (idempotent)
PYTHONPATH=src python3 scripts/assemble_paper3.py --verdict research/_verdict.md

# 9. VERIFICATION: recompute every quoted quantity from raw artefacts
PYTHONPATH=src python3 scripts/verify_paper3.py \
   --runs FR_flat,FR_flat1b,FR_A,FR_B,FR_flat_g4,FR_flat_g16,FR_A_g16,FR_flat_long,FR_A_long
```

Scripts added for this work: `src/hmoe/hier.py` (models), `scripts/train_hier.py`, `scripts/matched_hier.py`, `scripts/hier_init_blowup.py`, `scripts/hier_routing_metrics.py`, `scripts/hier_difficulty_curve.py`, `scripts/oracle_hier.py`, `scripts/slot_degeneracy.py`, `scripts/gen_divergence.py`, `scripts/composition_depth.py`, `scripts/hier_matched_steps.py`, `scripts/hier_tables.py`, `scripts/measure_all_hier.sh`, `scripts/final_measure.sh`, `scripts/wave2.sh`, `scripts/wave3.sh`, `scripts/granularity_ladder.py`, `scripts/paired_compare.py` (McNemar), `scripts/assemble_paper3.py`, `scripts/verify_paper3.py`.

Nothing in `research/PAPER_1_matched_cost.md` or `research/FINDINGS_AND_THEORY.md` was edited by this work; Condition 2 is cited from them and was not rerun.